



BASE24[®]

BIC ANSI Interface Manual

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Print Record

02/94, Version 1 (new)

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Preface

This manual describes the BIC ANSI Interface process, which is used to establish a communications link between a BASE24 network and one other co-network. This co-network can be another BASE24 network, but can also be any transaction processing network that conforms to the standards set forth in this manual for message formats and processing.

Audience

This manual is intended as a reference source for those persons responsible for the interface between BASE24 and a co-network. It provides a comprehensive description of all the functions performed by the BIC ANSI Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.

Prerequisites

This manual describes the BIC Interface process that uses the American National Standards Institute (ANSI) X9.2–1980 message format. A description of the BIC Interface process that uses the International Organization for Standardization (ISO) message format is included in the ***BASE24 BIC ISO Interface Manual*** and ***BASE24 BIC ISO Standards Manual***. It may be helpful to become familiar with the BIC Standard External Message (SEM) documented in appendix A of this manual.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for

readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Files Maintenance Manual*** discusses the Base and interchange files used with the BIC ANSI Interface.
- The ***BASE24 Logical Network Configuration File Manual*** discusses in detail the assigns and params used in configuring the BIC ANSI Interface.
- The ***BASE24 Log Message Reference Manual*** discusses in detail how to access log messages generated by the BIC ANSI Interface.
- The ***BASE24 Network Environment Source File Manual*** discusses in detail the process, station, line and device declarations used in configuring the BIC ANSI Interface.
- The ***BASE24 Text Command Reference Manual*** describes the each text command sent via the network control facility that can be issued for the BIC ANSI Interface. This manual includes the corresponding command syntax used with each command.
- The ***BASE24-atm Transaction Processing Manual*** discusses in detail the Standard Internal Message (STM) used by the BIC ANSI Interface.
- The ***BASE24-pos Settlement and Reporting Manual*** discusses in detail BASE24-pos settlement, cutover, balancing, and reporting.
- The ***BASE24-pos Transaction Processing Manual*** discusses in detail the POS Standard Internal Message (PSTM) used by the BIC ANSI Interface.
- The ***BASE24-pos Transaction Security Manual*** discusses in detail PINs, Message Authentication Codes (MACs), and dynamic key management used to set up transaction security.

Software Release

This manual documents standard processing for Release 5.0 of BASE24-atm and BASE24-pos.

Customer-specific code modifications (i.e., RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The ***BASE24 BIC ANSI Interface Manual*** is divided into the following sections and appendixes:

Section 1, “Introduction,” provides an overview of the BIC ANSI Interface process and its functionality.

Section 2, “Configuration and Control,” provides descriptions of the configuration and control parameters used by the BIC ANSI Interface. This includes the guidelines for setting up control files, configuring co-networks and stations, and configuring timers.

Section 3, “Network Management and Operations,” provides information on network management messages. These include logon, logoff, echo test, and dynamic key management messages.

Section 4, “Transaction Handling,” describes how the BIC ANSI Interface handles various types of transactions.

Section 5, “Cutover and Reporting,” describes cutover processing and report generation.

Appendix A, “BIC ANSI Interface External Message,” describes the BIC ANSI external message.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (e.g., SW-DS010-01 for the ***BASE24 BIC ANSI Interface Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (e.g., 02/94 for February, 1994) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions. The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the

changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Section 1

Introduction

This section introduces the BIC ANSI Interface process and provides an overview of BIC ANSI Interface functionality. This section also identifies the BASE24 files used by the BIC ANSI Interface. It includes the information on the following topics:

- Supported transaction types
- Message processing
- BASE24 file usage
- Interchange reporting
- PIN encryption
- Message formats
- Log message generation
- Data communications lines
- Interchange Log File usage
- Address verification
- BIC network usage

The BASE24 Interchange

The BASE24 Interchange, or BIC, serves as a link between two networks. It allows a cardholder of one network to initiate a transaction at a terminal belonging to the other network.

BIC ANSI connects a BASE24-atm or BASE24-pos network with a *co-network*—another BASE24 network or another transaction processing network which conforms to BIC ANSI standards. These standards include sending and accepting the appropriate form of the BIC ANSI external message and providing the right message flows to effect processing.

Although the co-network may be another BASE24 network, for the purposes of this manual the co-network is considered to be an unknown.

The BIC ANSI Interface Process

The BIC ANSI Interface process is responsible for maintaining the relationship between BASE24 and the co-network. In order to maintain that relationship, the BIC ANSI Interface process must perform a variety of functions. These functions include the following:

- Facilitating message exchange between BASE24 and the co-network.

When BASE24 receives a transaction from the co-network, the BIC ANSI Interface translates the message into the appropriate internal message and sends the message to a BASE24 Authorization process. When BASE24 receives a transaction for the co-network to authorize, the BIC ANSI Interface translates the message from the internal message format to the external message format and forwards the message to the co-network.

- Providing store-and-forward capabilities.

When the link between BASE24 and the co-network is down, the BIC ANSI Interface stores messages intended for the co-network in its Store-and-Forward File (SAF). When the link is restored, the messages are retrieved from the file and sent to the co-network.

- Facilitating settlement between BASE24 and the co-network.

The BIC ANSI Interface processes transactions based on its own reporting day. And, since the co-network cutover rarely occurs at the same time as the BASE24-atm and/or BASE24-pos cutover, the reporting day and the transactions to be settled will differ between the two. The BIC ANSI Interface produces a set of interchange reports to reconcile these points of view. These reports allow the BASE24 participants to reconcile the settlement positions of the co-network and BASE24.

- Managing communications between BASE24 and the co-network.

The BIC ANSI Interface sends and receives network management messages to determine the status of the link between BASE24 and the co-network. Network management messages include logon, logoff, echo test, and dynamic key management messages.

Supported Transaction Types

The BIC ANSI Interface process provides support for both BASE24-atm and BASE24-pos transactions. In addition, the BIC ANSI Interface functions on behalf of both issuers and acquirers.

An issuer processes transactions on behalf of the card-issuing financial institution. In the case of the BIC ANSI Interchange, this means that the BIC ANSI Interface process receives transactions from the co-network that require authorization by a BASE24 authorization process.

An acquirer accepts transaction requests that must be sent to another network for authorization. In the case of the BIC ANSI Interchange, this means that the BIC ANSI Interface will forward transactions that were initiated on the BASE24 network to the co-network for authorization.

Different transaction sets are supported by the BIC ANSI Interface depending on the BASE24 products installed. A supported transaction can be received from or sent to the co-network for authorization. The lists below identify the supported transactions.

Supported BASE24-atm Transactions

If BASE24-atm is installed, the following transactions are supported by the BIC ANSI Interface:

- Balance inquiries
- Check guarantee
- Check verification
- Deposits
- Messages to financial institutions
- Payments enclosed
- Payments from/to
- Transfers
- Withdrawals

These messages are supported for checking, savings, and credit accounts.

Supported BASE24-pos Transactions

If BASE24-pos is installed, the following messages are supported by the BIC ANSI Interface:

- Balance inquiries
- Card verifications
- Cash advances
- Cash advance adjustments
- Check guarantee
- Check verification
- Mail/phone orders
- Merchandise return adjustments
- Merchandise returns
- Normal purchases
- Preauthorization completions
- Preauthorizations
- Purchase adjustments
- Purchases with cash back

These messages are supported for checking, savings, NOW, and credit accounts.

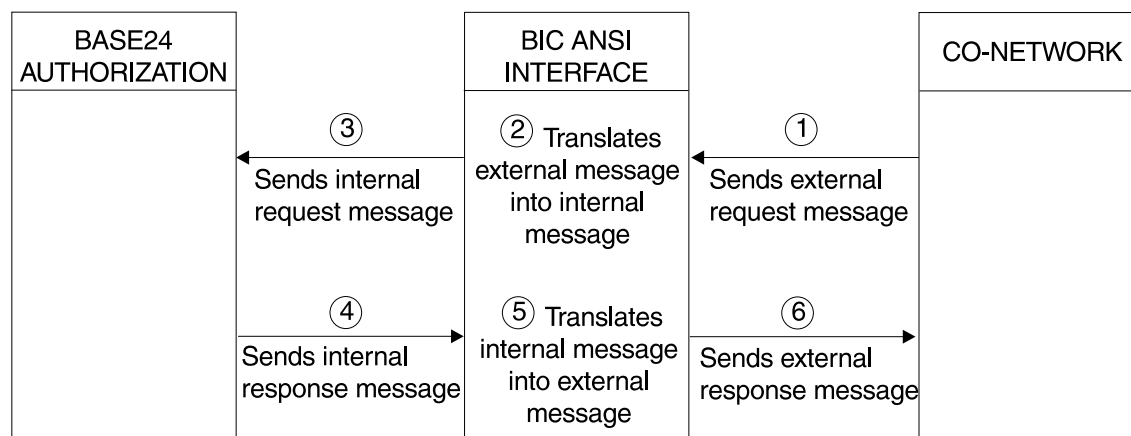
If both BASE24-atm and BASE24-pos are present, all of the transactions listed above are supported.

Message Processing

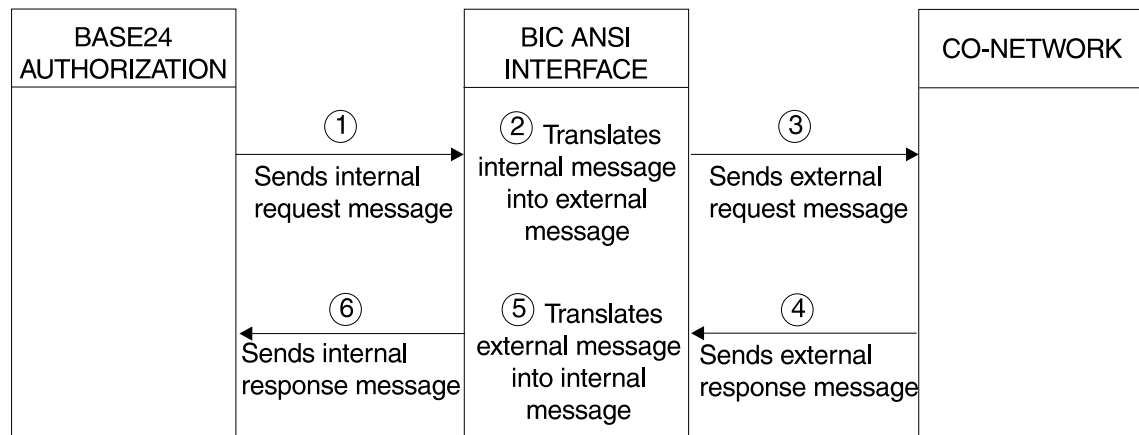
The following subsections provide an overview of transaction flows for normal and link-down processing. They also identify the message set (different than the transaction set) that is supported by the BIC ANSI Interface.

Normal Message Flow

When a message from the co-network is received by the BIC ANSI Interface process, it is translated from the external message into the appropriate internal message format. The internal message is then sent to the appropriate BASE24 Authorization process. When the message has been processed, the authorizing process sends an internal response back to the BIC ANSI Interface. Using information from the internal message, the BIC ANSI Interface formats an external response message and sends it back to the co-network. If completion messages are required, the co-network returns a completion message to the BIC ANSI Interface. In turn, the completion message may require a completion acknowledgment be sent to the co-network. This message flow is illustrated below.



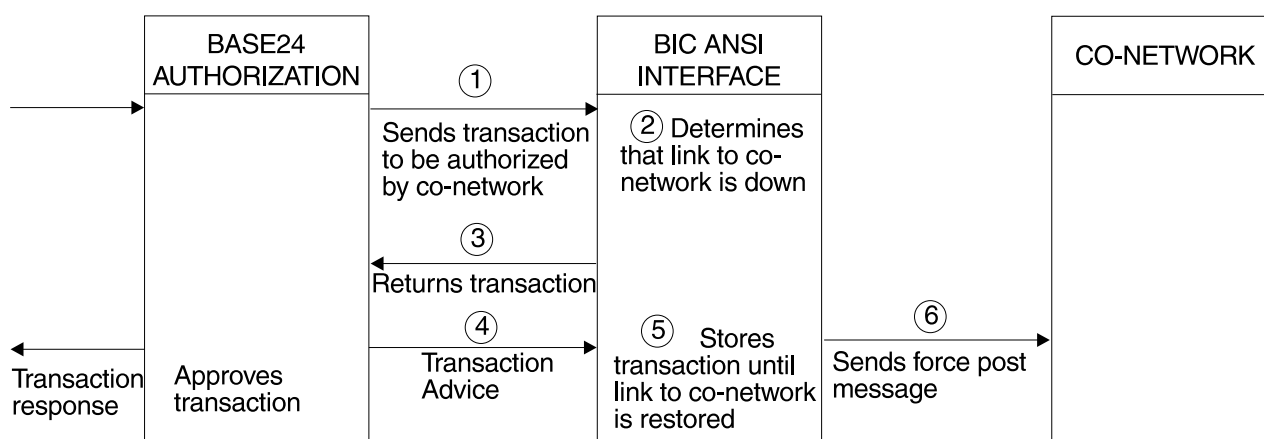
When a message from a BASE24 Authorization process is received by the BIC ANSI Interface process, it is translated from the internal message into the external message format. The external message is then sent to the co-network to be authorized. When the message has been processed, the co-network sends an external response back to the BIC ANSI Interface. Using information from the external message, the BIC ANSI Interface formats an internal response message and sends it back to the BASE24 Authorization process. If completion messages are required, the BIC ANSI Interface returns a completion message to the co-network. In turn, the completion message may require a completion acknowledgment be sent to the BIC ANSI Interface. This message flow is illustrated below.



Link-Down Message Flow

If stand-in authorization is allowed, the acquirer network has the option of authorizing requests when the link to the issuer network is down. The BIC ANSI Interface returns the request message to the acquirer—either a BASE24 Authorization process or the co-network—as a failed request needing offline authorization. If the transaction is approved, a force post message is forwarded to the issuer network when the link is reestablished. This message flow is illustrated below.

Note: Stand-in authorization is configured for the BIC ANSI Interface in the STAND-IN AUTH field on screen 12 of the Interchange Configuration File (ICF). If the STAND-IN AUTH field is set to 1, stand-in authorization is allowed for the specified BIC ANSI Interface process. If the STAND-IN AUTH field is set to 0, stand-in authorization is not allowed for the specified BIC ANSI Interface process.



If stand-in authorization is not allowed and an alternate destination is available, the link-down situation is handled by using alternate routing to send the transaction to the alternate destination. However, if stand-in authorization is not allowed and an alternate destination is not available, the BIC ANSI Interface responds to a link-down situation by denying all transactions until the link is reestablished.

Supported Messages Types

The BIC ANSI Interface supports the following internal and external message types:

Type	Description	Internal	External
0200	Financial Transaction Request	✓	✓
0210	Financial Transaction Request Response	✓	✓
0220	Financial Transaction Advice	✓	✓
0221	Financial Transaction Advice Repeat		✓
0230	Financial Transaction Advice Response		✓
0400	Reversal Request		✓
0402	Card Issuer Reversal Request	✓	✓
0410	Reversal Request Acknowledgment		✓
0412	Card Issuer Reversal Request Response		✓
0420	Acquirer Reversal Advice	✓	
0500	Acquirer Reconciliation Request		✓
0510	Acquirer Reconciliation Request Response		✓
0700	Inquiry Request		✓
0700	Inquiry Request Response		✓
0800	Network Management Request		✓
5400	Adjustment	✓	

Network Management Messages

Network management messages are used to manage the operational status of the communications lines between BASE24 and the co-network or to exchange PIN encryption keys between BASE24 and the co-network (that is, 0800 messages for new key, change key, verify key, or repeat key).

BASE24 supports four types of network management messages: logon messages, echo test messages, logoff messages, and dynamic key management messages. These messages are sent as 0800 messages and require 0800 messages in response.

BASE24 Files

The BIC ANSI Interface process uses a number of files. These files, which can be divided into control files and transaction files, are identified below.

Control Files

Control files contain configuration information and parameters that determine how the BIC ANSI Interface process performs certain functions. The control files used by the BIC ANSI Interface include the following:

- Interchange Configuration File (ICF)
- Key File (KEYF)
- Network Environment Source File (NEFS)
- Logical Network Configuration File (LCONF)

For more information on these files and how they are used by the BIC ANSI Interface process, see section 2.

Transaction Files

Transaction files are used to store information about individual transactions processed through the BIC ANSI Interface. The transaction files used by the BIC ANSI Interface include the following:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)

For more information on these files and what they contain, you can use DDLDOC. DDLDOC is a program that allows you to print out your DDL record structures in TAL or COBOL. For more information on how to use DDLDOC, see the ***BASE24 Files and Record Structures Reference Manual***.

Interchange Reporting

The BIC ANSI Interface produces two reports: the SETL01-BIC and the STAT09-BIC. The purpose of these reports is to facilitate settlement between BASE24 and the co-network. For more information on these reports, refer to section 5.

PIN Encryption

The BIC ANSI Interface process supports three PIN encryption options. These options are as follows:

- Clear PIN processing. All PINs handled by the co-network are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by the co-network are encrypted under the secure keys of a security module. The security modules that can be used include the Atalla Module MN10151, Atalla Network Security Processor (NSP), and the VISA Security Module.
- Encrypted PINs—software DES PIN processing. All PINs handled by the co-network are encrypted via software, using the DES algorithm. No security module is in use.

The specific PIN encryption method to use is determined by settings in the Key File (KEYF). The ***BASE24 Transaction Security Manual*** describes the KEYF settings required to support each of the options above.

Message Formats

The BIC ANSI Interface process routes messages using two different formats—internal or external.

The internal message format is used whenever information is routed between the BIC ANSI Interface process and the rest of BASE24. Depending on the BASE24 product receiving the message, the internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM) or the BASE24-pos Standard Internal Message (PSTM).

The external message format is used whenever information is routed between the BIC ANSI Interface process and the co-network. A message that is sent in the external message format is commonly referred to as a Standard External Message (SEM).

For more information on the STM or PSTM, see the BASE24-atm or BASE24-pos transaction processing manual. The SEM is described in appendix A of this manual.

Log Message Generation

The BIC ANSI Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system, BASE24, BIC, the security devices, and the associated lines and stations. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

ACI has developed a method that allows customers to access log message documentation electronically. Log messages are contained in files on a designated Tandem volume and subvolume. Each file contains log messages for a specific BASE24 process. Files on the designated Tandem volume and subvolume are organized according to product module ID (PMID). One PMID exists for every process in BASE24. PMIDs identify a specific process in the BASE24 system (e.g., SWBIC identifies the BIC ANSI Interface process). Providing log message documentation in files on the Tandem gives customers greater flexibility by allowing them to print out log message documentation as needed at their location or to electronically manipulate the data to suit their needs.

Documentation is available in file form for the log messages generated by the BIC ANSI Interface process. It is located on the BA50LOGM subvolume on the Tandem in a file named SWBIC. For a complete explanation of log message documentation, refer to the ***BASE24 Log Message Reference Manual***.

Data Communication Lines

The BIC ANSI Interface supports point-to-point and multipoint data communications lines between BASE24 and the co-network. Section 2 contains more information on configuring the data communications lines used by the BIC ANSI Interface process.

Interchange Log File (ILF)

The Interchange Log File (ILF) contains one record for each transaction processed by the BIC ANSI Interface. The ILF is used as input to the settlement report programs for the BIC ANSI Interface. These report programs read ILFs to get the transaction information they need to produce their reports. A new ILF is created each business day when cutover occurs. The business day reflected by the ILF is that of the co-network rather than BASE24.

The BIC ANSI Interface creates and writes a log record once for each transaction when the transaction response is available. If a reversal occurs, the original transaction is updated in place. It should be noted that external message logging is configurable based on the BIC-SEM-LOGGING param in the Logical Network Configuration File (LCONF). The BIC-SEM-LOGGING param sets a code indicating whether the Standard External Message (SEM) can be logged to the Interchange Log File (ILF). If the SEM cannot be logged to the ILF, the ILF is reduced in size by 544 bytes per record. Financial transactions log external messages if this param is set to Y (Yes). External messages are not logged if the value is set to N (No).

Customers can print out the basic record structure for the BIC ILF by using ACI's DDLDOC. For more information on DDLDOC, refer to the ***BASE24 Files and Record Structures Reference Manual***.

Address Verification

The BIC ANSI Interface supports BASE24-pos Address Verification processing.

Data elements 11 and 12 can be included in the SEM for BASE24-pos message types. These elements contain address information and verification results. The presence of these data elements in the SEM is determined by the setting of the BIC-VERSION-NUM param in the LCONF.

Section 2 contains valid LCONF param settings. For more information on address verification processing, see the ***BASE24-pos Address Verification Manual***.

Dynamic Key Management

The BIC ANSI Interface uses working keys to protect transaction data moving between networks. Working keys are used to maintain the proper security to protect personal identification numbers (PINs). Working keys should be changed regularly to ensure security. The monitoring and changing of these keys is known as key management and can be performed manually using text commands or automatically using dynamic key management.

Dynamic key management allows co-networks to automatically change working keys based on thresholds set for time, number of transactions, and number of single or consecutive errors. One or more working keys can be changed when any one of the defined thresholds is surpassed.

Dynamic key management requires the use of hardware security modules to perform PIN encryption and PIN verification. It is not supported when these functions are performed in software.

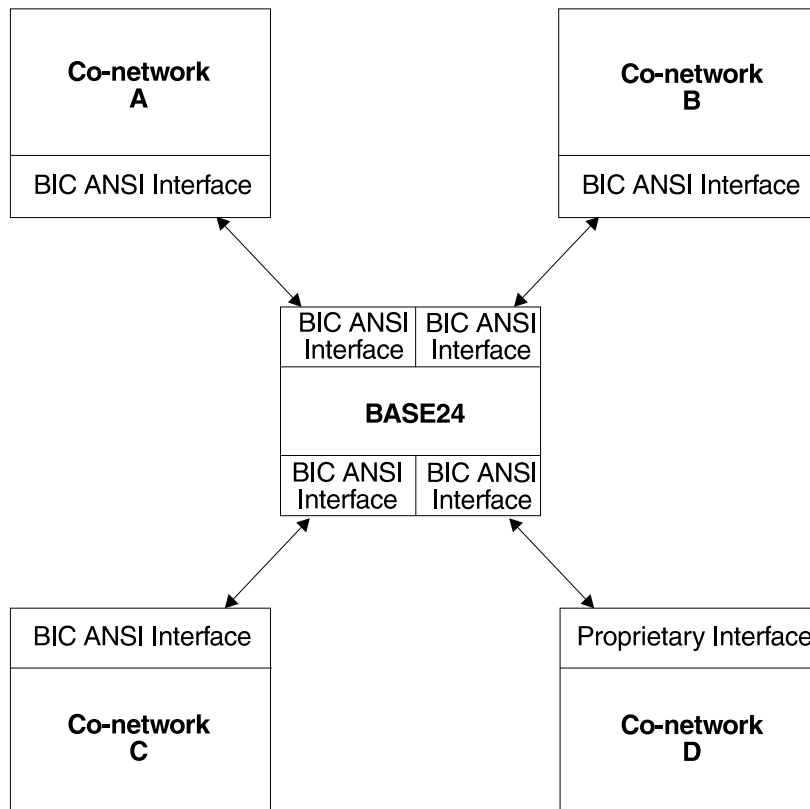
Fields in the Key File (KEYF) must be set to specific values to allow for dynamic key management. See section 2 for a listing of KEYF settings.

For more information about dynamic key management, refer to the ***BASE24 Transaction Security Manual***.

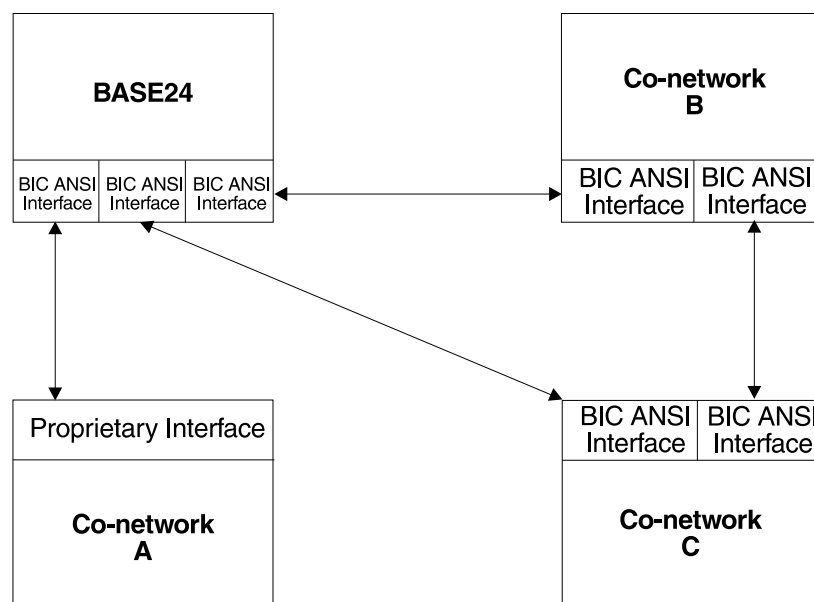
BIC ANSI Networks

The purpose of this manual is to describe the connection between a BASE24 system and a single co-network. However, it is possible to connect a BASE24 system to multiple co-networks. This type of system requires a separate BIC ANSI Interface process for each co-network to which the BASE24 system is connected.

A BIC ANSI network can be configured to require that all transactions be routed through a single member of the BIC ANSI network. The example illustrated below has a primary BASE24 system configured as the focal point for exchanging transactions within the entire BIC ANSI network. In this type of arrangement, the primary BASE24 system has a separate BIC ANSI Interface process for each co-network to which it is linked. Co-networks A, B, and C are other BASE24 systems and each requires one BIC ANSI Interface process. Co-network D is a non-BASE24 system and requires a proprietary interface that conforms to BIC ANSI Interface standards for message formats and processing.



A BIC ANSI network can also be configured to allow for certain transactions to be exchanged directly between members of the BIC ANSI network. An example of this is illustrated below. In this illustration, the configuration consists of a primary network and three co-networks, two of which are other BASE24 systems. The primary BASE24 system can exchange transactions with each of the three co-networks; co-networks B and C, which are also BASE24 systems, are configured to exchange transactions directly with each other. Co-network A is a non-BASE24 system. In this type of arrangement, the primary BASE24 network requires three BIC ANSI Interface processes, co-networks B and C each require two BIC ANSI Interface processes, and co-network A requires a proprietary interface that conforms to BIC ANSI Interface standards for message formats and processing.



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Section 2

Configuration and Control

This section lists the steps required to install the BIC ANSI Interface. In addition, it provides guidelines for setting up the following required control files:

- Logical Network Configuration File (LCONF)
- Interchange Configuration File (ICF)
- Key File (KEYF)

LCONF Params and Assigns

The BIC ANSI Interface process reads the LCONF, whenever it is initialized or warmbooted, for the assigns and params it requires during processing. *Assigns* associate the physical names of objects in the logical network to names used in BASE24 code. Assign information must be present, otherwise BASE24 is unable to process. Most assigns link the actual physical file names to logical file names as they are referenced in BASE24 program code. For example, in order to access the Interchange Configuration File (ICF), a process or server must be able to identify the fully-qualified name of the file as it exists in a logical network. The LCONF must contain an assign linking the ICF name to the actual physical file. Other assigns link the physical names for different hardware to the names as they are referenced in the BASE24 program code. For example, the types of hardware referenced in BASE24 program code includes spooler locations, terminal locations, and security devices.

Params set processing values at the logical network level. In some instances when param information is not present, a process abends during initialization; however, some params are not critical for processing.

The assigns and params used by the BIC ANSI Interface process are described below. Some of the assigns and params are required, and the BIC ANSI Interface process abends if a required assign or param is missing or invalid. The remaining assigns and params are optional, and the BIC ANSI Interface process uses the default values given if an optional assign or param is missing or invalid.

Assigns

The BIC ANSI Interface requires several assigns in the LCONF in order to perform processing. These assigns are as follows:

ICF — Identifies the name of the Interchange Configuration File (ICF).

ILF — Identifies the name of the Interchange Log File (ILF).

LMCF — Identifies the name of the Log Message Configuration File (LMCF).

KEYF — Identifies the name of the Key File (KEYF).

PLF — Identifies the name of the PCS Log File (PLF) and indicates BASE24-pcs is being used to monitor the network. If Logger is being used as the network logging facility and if the BIC ANSI Interface finds this assign in the LCONF, the BIC ANSI Interface sends any log messages to the BASE24-pcs system, in addition to sending the messages to Logger.

If Tandem's Event Management Service (EMS) is being used as the network logging facility, BIC ANSI Interface messages are automatically filtered to the BASE24-pcs system.

RPT — Identifies the name of the Report Perusal File (RPT). This assign is required if reports are to be run.

SDF — Identifies the name of the Switch Dispute File (SDF).

SECURE-DEV1 — The name of the primary security module used for PIN management, message authentication processing, and security module key management. For security devices other than the Atalla Network Security Processor (NSP), this assign specifies the Tandem SYSGEN device name (e.g., \B24.\$SECDVA). For Atalla NSP security devices, this assign specifies the PPD name given to the NSP interface module used with the security device (e.g., \B24.\$ASM1). The device name or interface module name must be compatible with Tandem's Expand product.

SECURE-DEV2 — The name of the secondary security module used for PIN management, message authentication processing, and security module key management. For security devices other than the Atalla Network Security Processor (NSP), this assign specifies the Tandem SYSGEN device name (e.g., \B24.\$SECDVA). For Atalla NSP security devices, this assign specifies the PPD name given to the NSP interface module used with the security device (e.g., \B24.\$ASM1). The device name or interface module name must be compatible with Tandem's Expand product.

STF — Identifies the name of the Switch Terminal File (STF).

SW-RPT01-PROG-NAME

SW-RPT09-PROG-NAME — Identify the names of the interchange report programs. These entries are required if these reports are to be generated.

RPT-ERROR-FILE

SW-RPT-OBEY-FILE — Identify the name of the spooler location to receive report error messages and the name of the obey file used to start the interchange reports. These entries are required if reports are to be generated.

Params

Several params are used by the BIC ANSI Interface during processing. These params are as follows.

ACCT-NUM-TYP — A code indicating whether account numbers are stored in binary or display form. The binary form allows the account number to be up to 28 digits in length. The display form allows the account number to be up to 19 digits in length. Valid values are as follows:

- D = Display format (default)
- B = Binary format

ATM-LN-CUTOVER — The time of day all BASE24-atm processes in the entire logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59. This param is required for BASE24-atm.

BASE24-REL — A code indicating whether the BASE24 logical network is currently running Release 3.4/4.0 or Release 5.0. The Release 5.0 BIC ANSI Interface process uses this param to determine whether to use Release 5.0 or previous release processing parameters.

This param must be set to 50 if the BASE24 network being used is Release 5.0.

If the BASE24-REL param contains a value other than 50 or is not present in the LCONF, the BIC ANSI Interface process assumes the network running is a previous release BASE24 network (i.e., Release 3.4/4.0 BASE24 network) and performs according to the processing parameters of the previous release.

The BASE24-REL param is the last param the BIC ANSI Interface process retrieves. The BIC ANSI Interface process retrieves the LCONF file assigns according to how this param is configured.

BIC-ILF-NOTIFY-INTERVAL — The number of writes made to the ILF between calculating the percentage of file full. Each time a write is made to the ILF, a counter is increased by one. When the counter is equal to the value specified in this param, The BIC ANSI Interface process performs a calculation to determine how full the ILF is. The counter is then reset to zero. When the file reaches the percent full specified in the BIC-ILF-NOTIFY-PERCENTAGE param, the BIC ANSI Interface process logs a message indicating the file is becoming full. Until more file space is made available, the BIC ANSI Interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10–32766. The default value is 50.

BIC-ILF-NOTIFY-PERCENTAGE — The percentage of file full at which the BIC ANSI Interface process logs a message indicating the Interchange Log File (ILF) is becoming full. The BIC ANSI Interface process calculates the percentage of file full and continues to log messages at the interval indicated in the BIC-ILF-NOTIFY-INTERVAL param until more file space is made available as described in the following paragraphs.

Each time the BIC ANSI Interface process writes the number of transactions log file records specified in the BIC-ILF-NOTIFY-INTERVAL param, it checks the ILF to determine how full it is. The calculation to determine how full the file is assumes that the record length is fixed, rather than variable. Because the records can be variable length and may be shorter than the maximum length assumed by the calculation, the result of this check is always a percentage of file used that is greater than or equal to the actual percentage of file space used.

If the calculation results in the percentage specified in the BIC-ILF-NOTIFY-PERCENTAGE param, or more, of its capacity, the BIC ANSI Interface process generates a log message each time it writes the number of records specified in the BIC-ILF-NOTIFY-INTERVAL param and as long as the transaction log file calculation results in the percentage specified in the BIC-ILF-NOTIFY-PERCENTAGE param, or above, of its capacity.

Depending on the number and size of tokens logged to the file with the internal message, this calculation and the resulting log messages may not accurately reflect an interchange log file approaching capacity. Valid values are 10–99. The default value is 70.

BIC-SAF-NOTIFY-INTERVAL — The number of writes made to the Store-and-Forward File (SAF) between calculating the percentage of file full. Each time a write is made to the SAF, a counter is increased by one. When the counter is equal to the value specified in this param, the BIC ANSI Interface process calculates how full the SAF is. The counter is then reset to zero. When the file reaches the

percent full specified in the BIC-SAF-NOTIFY-PERCENTAGE param, the BIC ANSI Interface process logs a message indicating the file is becoming full. Until more file space is made available, the BIC ANSI Interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10–32766. The default value is 50.

BIC-SAF-NOTIFY-PERCENTAGE — The percentage of file full at which the BIC ANSI Interface process logs a message indicating the Store-and-Forward File (SAF) is becoming full. The BIC ANSI Interface process continues to log messages at the interval indicated in the BIC-SAF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10–99. The default value is 70.

BIC-SEM-LOGGING — A flag indicating whether the Standard External Message (SEM) can be logged to the ILF. If SEM logging is not permitted, the ILF is reduced in size by 544 bytes per record. Valid values are as follows:

Y = Yes, SEM logging is permitted (default).
N = No, SEM logging is not permitted.

BIC-VERSION-NUM — The version of the BIC ANSI Interface being used. Both co-networks must be using the same version. The version of the BIC ANSI Interface impacts the message formats accepted. Valid values are as follows:

00000100 = Previous version
00000200 = Current version (default)
00000300 = Current version with address verification

DATE-DISPLAY-TYPE — A code indicating the format of the month and day portion of the dates on reports and BASE24-atm Device Handler receipts. When dates include the year, the year portion is not affected by this param. Valid values are as follows:

1 = Day before month (e.g., DDMM or DDMMM)
2 = Month before day (e.g., MMDD or MMMDD) (default)

MAX-STF-RECORDS — The maximum number of terminals defined in the Switch Terminal File (STF) for a BIC ANSI Interface process.

POS-LN-CUTOVER — The time of day all BASE24-pos processes in the entire logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59. This param is required for BASE24-pos.

SECURE-DEV-CONF-CMD100 — Specifies the command 100 text sent to Atalla Network Security Processors (NSPs) when the ANSI Host Interface is initialized or warmbooted.

SECURE-DEV-CONF-TXT — Specifies the configuration text sent to Atalla NSPs when the ANSI Host Interface is initialized or warmbooted.

SECURE-DEV-FAIL-NOTIFY — Specifies the interval, in number of transactions, at which the ANSI Host Interface logs messages once a security module is determined to be inoperative.

SECURE-DEV-MONITOR-THRESHOLD — Specifies the interval, in number of transactions processed by the security module defined in the SECURE-DEV1 assign, at which the security module defined in the SECURE-DEV2 assign is checked.

SECURE-DEV-TIM-LMT — Specifies the number of seconds the ANSI Host Interface waits for a response from the security module.

SECURE-DEV-TYP — Specifies the type of security modules named in the SECURE-DEV1 and SECURE-DEV2 assigns.

SW-ILF-FILE-RETENTION — The number of days worth of Interchange Log Files (ILFs), prior to the current business day's ILF, which are to remain on the system. After the specified number of days, the BIC ANSI Interface process deletes the ILF from the system. For example, if the value in this param is 3 and the current business date is 93/11/05, ILFs dated 93/11/05 and 93/11/04 are on the system and open. ILFs dated 93/11/03 and 93/11/02 are on the system and closed. The ILFs dated 93/11/01 and earlier have been deleted from the system by the BIC ANSI Interface process. Valid values are 3–99. The default value is 7.

SW-RPT-STARTUP — The time the BIC ANSI Interface process runs its report obey file. Valid values, in military time, are from 00:00 (12:00 a.m.) to 23:59 (11:59 p.m.). The default value is ATM-LN-CUTOVER value or POS-LN-CUTOVER value, whichever one contains the latest time.

See the ***BASE24 Logical Network Configuration File Manual*** for more information on assigns and params that are not specific to the BIC ANSI Interface process, and for instructions on adding them to the LCONF.

Configuring a Co-Network

Each co-network with which BIC ANSI Interface communicates requires a record in the Interchange Configuration File (ICF). Co-networks are assigned stations and tied to the BIC ANSI Interface processes with which they communicate via the ICF.

A number of processing options are controlled at the co-network level, which enable institutions to set up co-networks with individual processing requirements. The majority of processing options available at the co-network level are stored in the ICF. Other options are specified in the KEYF. The processing options controlled at the co-network level are described below.

Specifying Timer Lengths

The BIC ANSI Interface process uses internal timers to control co-network message traffic. The length of each of these timers is specified in the ICF or KEYF. Some timer intervals are specified once at the interface level, while others are specified at the interface and product level (e.g., you can specify one time limit for inbound BASE24-atm messages and a different time limit for inbound BASE24-pos messages).

The table below identifies each timer used by the BIC ANSI Interface, along with the level at which it is defined and the file in which it is defined. Each of these timers are discussed later in this section.

Timer	Interface Level	Interface/Product Level
Network Management Message timer	ICF	
Extended Network Management Message timer	ICF	
Store-and-Forward Message timer	ICF	
Wait-for-Traffic timer	ICF	
Performance timer	ICF	
Outbound Request timer		ICF
Inbound Request timer		ICF

Timer	Interface Level	Interface/Product Level
Completion Acknowledgment timer		ICF
PIN Key Exchange timer	KEYF	
Clear Old Key timer	KEYF	

Interspersing Store-and-Forward Messages

Store-and-forward messages are interspersed with request messages. Interspersing store-and-forward messages allows other transactions to be processed during store-and-forward activity, while receiving all store-and-forward messages prior to further requests may cause requests to be processed offline or be declined because of an inability to send them to the co-network.

In addition, messages sent to the SAF are *single-threaded*. Single-threaded means that only one message of any kind can be outstanding to a station at a time. In this case, the BIC ANSI Interface must receive a response or the message must time out before another message can be sent to the station.

The BIC ANSI Interface sends the following messages types directly to the SAF instead of to the co-network:

- Reversal messages
- Force post messages
- Adjustment messages
- Chargeback messages
- Cutover messages
- Administrative messages

Sending Text-Level Acknowledgments to a Co-Network

Acquiring institutions must specify whether the co-network expects text-level acknowledgments from the BIC ANSI Interface. If text-level acknowledgments are required, BASE24 returns a 0130 message for each 0120 and 0121 message; a 0230 message for each 0220 or 0221 message; a 0430 message for each 0420 or 0421 message; and a 0530 message for each 0520 or 0521 message.

Institutions indicate whether text-level acknowledgments are required via the ACK TO SWITCH field on ICF screen 1.

Expecting Text-Level Acknowledgments from a Co-Network

Issuer Institutions must indicate whether the BIC ANSI Interface should expect text-level acknowledgments. If text-level acknowledgments are required, BASE24 expects a 0130 message for each 0120 and 0121 message; a 0230 message for each 0220 or 0221 message; a 0430 message for each 0420 or 0421 message; and a 0530 message for each 0520 or 0521 message.

Institutions indicate whether they send text-level acknowledgments via the ACK FROM SWITCH field on ICF screen 1.

Note: The BIC ANSI Interface process always requires acknowledgments for its store-and-forward messages. If text-level acknowledgments are not to be sent, the BIC ANSI Interface process requests a *logical acknowledgment* from the BASE24 Nucleus. A logical acknowledgment notifies the BIC ANSI Interface process that the message has been sent successfully to the co-network.

If this field indicates that text-level acknowledgments are not required and the message was a store-and-forward message, the original message will not be available to resend to the co-network. This is because the message would have been deleted when the Nucleus returned a logical acknowledgment that the message was sent successfully.

Specifying Maximum Timeouts

Institutions must indicate the number of consecutive timeouts of non-network management and non-store-and-forward messages that can occur at a station before the station is marked down. For example, if 3 is specified, the BIC ANSI Interface process marks the station down upon the third consecutive timeout. However, if only two consecutive timeouts have occurred, the station is not marked down.

Institutions control when a station is marked down because of timeouts via the MAXIMUM TIMEOUTS field on ICF screen 1.

Note: Dynamic key management (DKM) timeouts do increase this count.

Enabling Network Management

Institutions specify whether the BIC ANSI Interface is allowed to initiate network management messages (that is, logon, logoff, echo test). This option can limit BASE24's ability to log on and solicit communications with a co-network.

Institutions indicate whether the BIC ANSI Interface can initiate network management messages via the NETWORK MANAGEMENT MESSAGE ENABLED field on ICF screen 3.

Note: This option only controls whether the BIC ANSI Interface can initiate the network management actions; the BIC ANSI Interface always responds to the network management messages it receives.

Also, although key management messages are sent as 0800 messages, they are not controlled by this option. The BIC ANSI Interface initiates key management messages based on settings in the Key File (KEYF). For more information on configuring the KEYF, refer to the ***BASE24 Transaction Security Manual***.

Specifying Maximum Store-and-Forward Retries

Institutions set the maximum number of times the BIC ANSI Interface process attempts to send a store-and-forward message to a co-network before sending that message to the log and deleting it from the SAF.

This option can be used to stop processing a message when, due to the number of timeouts on a store-and-forward message, it becomes apparent that the co-network cannot process the message. Institutions control this option via the MAX SAF RETRY field on ICF screen 3.

Specifying Maximum Outbound Messages

The BIC ANSI Interface allows institutions to specify the maximum number of request messages (that is, 0100, 0200, 0300, 0402, and 0500 messages) or advice messages (that is, 0120, 0220, 0320, 0420, and 0520 messages) that can be outstanding to an issuer co-network before the BIC ANSI Interface stops sending additional messages. Outstanding messages are those messages that have not been received (for example, a 0210 for a 0200 or a 0230 for a 0220).

This option allows the BIC ANSI Interface process to prevent processing when it becomes apparent that the co-network cannot process a message within the time limits required because of the number of messages already outstanding to the co-network.

Once the number of outstanding messages reaches this limit, the BIC ANSI Interface process marks any new request messages destined for the co-network as failed and returns them to their originating process. The BIC ANSI Interface process places any new advice messages destined for the co-network in the SAF. At this point, the co-network does not receive additional request or advice messages until the messages in the queue are either responded to or have timed out.

Institutions control the number of outstanding request and advice messages to a co-network via the OUTBOUND field on ICF screen 3.

Maximum Inbound Requests

The BIC ANSI Interface also provides a parameter for limiting the number of outstanding 0100 and 0200 messages to a BASE24-atm or BASE24-pos Authorization process. This parameter controls requests being sent from acquirer co-networks. Once the number of outstanding 0100 and 0200 messages reaches this limit, the BIC ANSI Interface process declines any new 0100 and 0200 messages coming from the co-network and returns them as 0110 and 0210 messages.

The number of outstanding requests to an Authorization process is controlled via the INBOUND field on ICF screen 3.

Note: This feature only affects 0100 and 0200 messages. Other message types are not included in the count of outstanding messages, nor is their processing affected in any way.

Data Prefix Characters

The BIC ANSI Interface allows institutions to define certain characters they want included in front of the messages they receive from the BIC ANSI Interface. An institution can specify up to nine characters to precede messages to their co-network system. Each time the BIC ANSI Interface process creates an external message to a co-network, it checks the ICF for data prefix characters. If the ICF contains data prefix characters for the co-network, the BIC ANSI Interface process places these characters at the front of the message.

Data prefix characters are placed in the ICF using the hexadecimal equivalents for the ASCII versions of the characters to be sent. For example, suppose a co-network required the following three characters in front of the message:

ESC 1 B
(escape)

To place these three values in an ICF record, a co-network would have to enter hexadecimal equivalents (six characters) for the ASCII representations of these values, which would be the following:

1B 31 42

These values would be stored in the ICF as six characters (1B3142) and would be converted back to the three ASCII characters at the time the message is created.

Because the BIC ANSI Interface does not support transparent data communications, protocol characters must not be used as data prefix characters. The following table shows some characters to avoid.

Hex	ASCII Display	Hex	ASCII Display
01	SOH	02	STX
03	EXT	04	EOT
05	ENQ	06	ACK
10	DLE	15	NAK
16	SYN	17	ETB
1F	ITB		

Authorization Process

The BIC ANSI Interface processes handling requests from acquirer networks determine the Authorization processes to which to route inbound transaction requests using the AUTH PROCESS field on ICF screen 8 for BASE24-atm or ICF screen 10 for BASE24-pos.

Setting Up Co-Network Stations

Stations serve as the links between BASE24 and a co-network. All messages sent to a co-network are actually sent to the stations defined for that co-network. Likewise, all messages received from a co-network are actually received from the stations defined to that co-network.

Up to 2 stations can be defined for each co-network. A station must first be defined as a co-network station in the Network Environment File (NEF). The station's symbolic name can then be placed in the ICF to fully define the station.

Station Types

Co-networks can define two types of stations for use by the BIC ANSI Interface process: send/receive stations and receive-only/no output stations. These stations are described below. Station types are defined via the TYPE fields on ICF screen 13.

Send/Receive Stations

Send/receive stations allow for both incoming and outgoing messages. The BIC ANSI Interface process can send messages to and receive messages from the co-network using this type of station.

Receive-Only/No Output Stations

Receive-only/no output stations only allow for messages coming in to BASE24. BASE24 never uses receive only/no output stations for sending messages to the co-network. The receive-only/no output station is used to avoid line connection when using a point-to-point or SNA protocol. In this case, communications are received on one station and sent from a different station.

Note: For single-threading, messages and their responses must be sent and received via the same stations. For example, if a request is sent to station A, the response to that request must also be returned via station A. The response cannot be returned via a different station. Therefore, receive-only/no output stations should not be defined for single-threaded co-networks.

Setting Up the ICF Station Table

Stations are numbered from left to right, top to bottom on the ICF file maintenance screens. This numbering scheme is of particular importance to co-networks with multiple stations on their lines.

When selecting a station to which to send a message, the BIC ANSI Interface process goes through its station table from start to finish until it finds a station available to receive the message.

If lines have multiple stations on them, the stations from each line should be alternated when setting up the ICF station table, and not numbered sequentially. Alternating the stations balances the transaction load so that if one station is chosen from one line, the next station chosen is on a different line.

Selecting Stations for Sending Messages

To select the station to which to send a message, the BIC ANSI Interface process begins with the first station in the ICF for that co-network and evaluates each station based on the following criteria:

1. It must be a send/receive station and the station must be marked up. The BIC ANSI Interface process never sends messages to receive only/no output stations.
2. Since the co-network is single-threaded, the station cannot be busy; that is it must not have a message pending.

The BIC ANSI Interface process sends the message to the first station that meets the above criteria. To send the next message, the BIC ANSI Interface process performs its station selection in the same manner, starting with the next station. For example, if the last message was sent to the first station, the BIC ANSI Interface would begin its station selection with the second station.

Configuring Timers

The BIC ANSI Interface process uses internal timers to control co-network message traffic. The length of each of these timers is specified in the ICF.

Network Management Message Timer

A Network Management Message timer is set each time an 0800 message is sent to a co-network station. The Network Management Message timer limits the amount of time that the BIC ANSI Interface process waits for a response after transmitting an 0800 message. The number of seconds to use for this timer is taken from the NETWORK MANAGEMENT field on ICF screen 3.

The actions taken when a Network Management Message timer expires depend on the type of message for which the timer was set. If the message was a logon, logoff, or echo test message, the BIC ANSI Interface performs as follows:

1. Marks the station down.
2. Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the BIC ANSI Interface process marks the co-network down.
3. Sets an Extended Network Management Message timer for the station.

If the 0800 message was a dynamic key management message, the BIC ANSI Interface performs as follows:

1. Adds one to the count of consecutive timeouts accumulated for the station.
2. Checks the number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, processing continues with step 3.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the BIC ANSI Interface process performs as follows:
 - 1) Marks the station down.
 - 2) Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, so the BIC ANSI Interface process marks the co-network down.

3. If the message that timed out was a repeat key or verify key message, the BIC ANSI Interface logs a message indicating that the message timed out
4. If the message that timed out was a new key or a change key message, the BIC ANSI Interface performs as follows:
 - a. Logs a message indicating that the message expired and that the message will automatically be retried.
 - b. Sets an Extended Network Management Message timer for the station. (The new key or change key message will be retried when this timer expires).
 - c. Sets a flag indicating a key change is in progress. No other key change other than the one currently in progress is allowed.

Extended Network Management Message Timer

An Extended Network Management Message timer is set each time a co-network station is marked down and each time a new key or change key message times out. The Extended Network Management Message timer specifies the following intervals:

- The amount of time that the BIC ANSI Interface waits after a station is marked down before attempting to reestablish communications with the station.
- The amount of time the BIC ANSI Interface waits after a new key or change key message times out before attempting to resend the message.

The number of seconds to use for this timer is taken from the EXTENDED NETWORK field on ICF screen 3.

If an Extended Network Management Message timer that was set when a station was marked down expires before any messages are received from the station, the BIC ANSI Interface process:

1. Sends an echo test message to the station.
2. Sets a Network Management Message timer for the echo test message.

If an Extended Network Management Message timer that was set after a new key or change key message time out expires, the BIC ANSI Interface determines whether there is a station available to which to send the message.

If a station is available, the BIC ANSI Interface sends the new key message or change key message to that station and sets a Network Management Message timer to wait for the response to the new key or change key message.

If no stations are available, the BIC ANSI Interface sets another Extended Network Management Message timer. When this timer expires, the BIC ANSI Interface attempts to send the message again.

For more information on how the BIC ANSI Interface selects a station for sending a message, see the topic “Selecting Stations for Sending Messages” earlier in this section.

Store-and-Forward Message Timer

A Store-and-Forward Message timer is set each time a store-and-forward message is sent to a co-network. The Store-and-Forward Message timer limits the amount of time that the BIC ANSI Interface waits for an acknowledgment after transmitting a store-and-forward message. The number of seconds to use for this timer is taken from the STORE AND FORWARD field on ICF screen 8 for BASE24-atm and ICF screen 10 for BASE24-pos.

The BIC ANSI Interface process only expects text-level acknowledgments to store-and-forward messages if the ACK FROM SWITCH field on ICF screen 3 contains the value Y. If text-level acknowledgments are not required from the co-network, the BIC ANSI Interface process expects a logical acknowledgment from the BASE24 Nucleus once the message has been sent successfully to the co-network. In this case, if the co-network rejects the message for any reason, the message will not be resent because it will have already been deleted from the SAF.

If this timer expires before an appropriate acknowledgment is received in response, the BIC ANSI Interface process:

1. Reads the message from the SAF.
2. Adds one to the SAF retries count for that record. The SAF retries count indicates the number of times the SAF record has been sent to the co-network. A separate count is kept for each SAF record.
3. Checks the record's SAF retries count against the maximum allowed by the MAX SAF RETRY field on ICF screen 3.

- a. If the SAF retries count is equal to the maximum allowed, the BIC ANSI Interface process performs as follows:
 - 1) Writes the SAF record to its log.
 - 2) Deletes the record from the SAF.
 - b. If the record's SAF retries count is less than the maximum allowed, the BIC ANSI Interface process continues with the next step.
4. Marks the SAF message to be resent.
5. Updates the message type, if necessary.

If the message type is 0120, 0220, 0320, 0420, or 0520, the BIC ANSI Interface process reads the corresponding record in the SAF and updates the message type in place to 0121, 0221, 0321, 0421, or 0521 respectively. Message types 0402 and 0412 are not changed.

Wait-for-Traffic Timer

A Wait-for-Traffic timer is set each time a station is marked up and each time a previously set Wait-for-Traffic timer expires. The Wait-for-Traffic timer specifies the amount of time that the BIC ANSI Interface waits for traffic on the line before sending an echo test message. The number of seconds to use for this timer is taken from the WAIT FOR TRAFFIC field on ICF screen 3.

The BIC ANSI Interface process sets Wait-for-Traffic timers for its send/receive stations only. Wait-for-Traffic timers are not set for receive-only/no output stations.

When this timer expires, the BIC ANSI Interface process determines whether there has been any message traffic to or from the station since the timer was set. Depending on whether there has been traffic on the line, the BIC ANSI Interface process performs as follows:

1. If there has been message traffic, the BIC ANSI Interface process simply sets another Wait-for-Traffic timer.
2. If there has not been message traffic, the BIC ANSI Interface process:
 - a. Sends an echo test message to the station.
 - b. Sets a Network Management Message timer for the echo test message.

Performance Timer

A Performance timer is set for each co-network at the start of that co-network's new performance statistics accumulation period. The Performance timer specifies the interval the BIC ANSI Interface uses for gathering performance statistics. The number of minutes to use for this timer is taken from the PERFORMANCE PERIOD field on ICF screen 3.

For each co-network, the BIC ANSI Interface process tracks the number of requests sent to the co-network, the number and percentage of requests that time out, and the average response time. These statistics are tracked on an interval basis and are available for the most current four intervals (interval zero through interval three with interval zero being the most current). The Performance timer triggers the end of one interval and the beginning of the next.

When a Performance timer expires, the BIC ANSI Interface process performs as follows:

1. Moves the performance statistics from interval two to interval three.
2. Moves the performance statistics from interval one to interval two.
3. Moves the performance statistics from interval zero to interval one.
4. Zeros out the performance statistics in interval zero to begin accumulating statistics again.
5. Sets a Performance timer for the co-network.

Note: These statistics are accessed via a Performance command entered from the Network Control Supervisor (NCS) screen. For information concerning the entry of the Performance command and the display of the statistics, please refer to the description of the Performance command in the ***BASE24 Text Command Reference Manual***.

Outbound Request Timer

An Outbound Request timer is set each time a 0100, 0200, 0205, or 0300 message is sent to a co-network. The Outbound Request timer limits the amount of time that the BIC ANSI Interface waits for a response from the co-network after transmitting a request message. The number of seconds to use for this timer is taken from the OUTBOUND field on the product-specific ICF screen.

If this timer expires before a response is received to the message, the BIC ANSI Interface process performs as follows:

1. Performs product-specific processing.
 - a. If the message is a BASE24-atm message, the BIC ANSI Interface process performs as follows:
 - 1) Retrieves the original message from memory.

The BIC ANSI Interface process retains a copy of the internal message for each message it sends to the co-network until it is notified that the message was received.
 - 2) Sets the STM to indicate the destination was not available. This identifies the 0200 message as a failure.
 - 3) Sets the STM to cause the Authorization process to generate a 0220 message if it subsequently stands in and authorizes the transaction.
 - 4) Sends the STM to the process where the message originated.

Note: The Outbound Request timer is also used for 0205 messages. If this timer expires before a 0215 message is received from the co-network, the BIC ANSI Interface process creates a 0215 with a response code of 72, declining the transaction.
 - 5) If the message is a BASE24-pos message, the BIC ANSI Interface process performs as follows:
 - a) Retrieves the 0200 message from memory (all 0100 messages are handled as 0200 messages internally).

The BIC ANSI Interface process retains a copy of the PSTM for each 0100 and 0200 message it sends to the co-network until it is notified that the message was received.
 - b) Sets the PSTM to indicate the destination was not available. This identifies the 0200 message as failed.

- c) Sets the PSTM to cause the Authorization module to generate a 0200 message if it subsequently stands in and authorizes the transaction.
 - d) Sends the PSTM to the process where the message originated.
- 2. Adds one to the count of consecutive timeouts accumulated for the station.
- 3. Adds one to the number of request timeouts for the co-network.

The BIC ANSI Interface process tracks this information as part of its co-network performance statistics.

- 4. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICF screen 3.
 - a. If the current number of consecutive timeouts is equal to the maximum allowed, no further action is taken.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the BIC ANSI Interface process:
 - 1) Marks the station down.
 - 2) Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the BIC ANSI Interface process marks the co-network down.
 - 3) Sets an Extended Network Management Message timer for the station.

Inbound Request Timer

An Inbound Request timer is set each time a 0100 or 0200 message is sent to a BASE24-atm or BASE24-pos Authorization process. The Inbound Request timer limits the amount of time that the BIC ANSI Interface waits for a response from a BASE24 Authorization process after transmitting a request message. The number of seconds to use for this timer is taken from the INBOUND field on the product-specific ICF screen.

If an Inbound Request timer expires before a response message is received, the BIC ANSI Interface process:

1. Performs product-specific processing.
 - a. If the message is a BASE24-atm message, the BIC ANSI Interface performs as follows:
 - 1) Retrieves the original message from memory.

The BIC ANSI Interface retains a copy of the internal message for each message it sends to the co-network until it is notified that the message was received.
 - 2) Updates the TYP field in the SEM.
 - b. If the TYP field is equal to 0700, the BIC ANSI Interface process changes it to 0710.
 - c. If the TYP field is equal to 0200, the BIC ANSI Interface process changes it to 0210.
 - 1) Sets the RESP-CDE.RESP field in the SEM to 54 to indicate that the transaction was denied because of an inability to process it.
 - 2) Sets the ROUTE.STAT field in the SEM to 11 to indicate that the destination was not available.
 - 3) Sends the SEM to the host.
 - d. If the message is a BASE24-pos message, the BIC ANSI Interface performs as follows:
 - 1) Sets the RESP-CDE.RESP field in the PSEM to 100, indicating that the transaction is declined because of an inability to process it.
 - 2) Sets the ROUTE.STAT field in the PSEM to 11, indicating the destination was not available.
 - 3) Sends the PSEM to the host.
2. Adds one to the number of request timeouts for BASE24.

Completion Acknowledgment Timer

A Completion Acknowledgment timer is set each time BASE24 sends a 0220, 0400, or 5400 message to a co-network. The Completion Acknowledgment timer limits the amount of time that BASE24 waits for an acknowledgment from the co-network after transmitting an advice message. The number of seconds to use for this timer is taken from the COMPLETION ACK field on the product-specific ICF screen.

The BIC ANSI Interface only expects text-level acknowledgments to completion messages if the ACK FROM SWITCH field on ICF screen 1 contains a value of Y. If text-level acknowledgments are not required from the interchange, the BIC ANSI Interface expects an acknowledgment from the BASE24 Nucleus instead. In this case, the Nucleus returns an acknowledgment once the message has been sent successfully to the interchange.

If a Completion Acknowledgment timer expires before an appropriate acknowledgment is received, the BIC ANSI Interface performs as follows:

1. Retrieves the original message from memory.

The BIC ANSI Interface retains a copy of the internal message for each message it sends to the co-network until it is notified that the message was received.
2. Changes the TYP field from 0220, 0400, or 5400 to 0221, 0401, or 5401, respectively.
3. Places the SEM in the Store-and-Forward File (SAF).
4. Adds one to the count of consecutive timeouts accumulated for the station.
5. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAX TIMEOUTS field on ICF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, no further action is taken.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the BIC ANSI Interface performs as follows:
 - 1) Marks the station down.
 - 2) Sets an Extended Network Management Message timer for the station.

6. Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the BIC ANSI Interface marks the co-network as down.
7. Adds one to the number of request timeouts for the co-network.
The BIC ANSI Interface tracks this information as part of its performance statistics.

Key Exchange Timer

Key Exchange timers are used to automatically initiate PIN key exchanges. If BASE24 is configured for dynamic key management, one Key Exchange timer is set for each PIN key being used by the BIC ANSI Interface. Key Exchange timers are set or reset at the following times:

- When the BIC ANSI Interface is initialized or warmbooted
- When the key for which the timer has been set is exchanged for any reason

The BIC ANSI Interface sets and uses Key Exchange timers for PIN keys when the PIN KEY TIMER VALUE field on KEYF screen 3 contains a nonzero value. PIN Key Exchange timers limit the amount of time that the BIC ANSI Interface uses a particular PIN key to encrypt and decrypt PINs sent between the BIC ANSI Interface and the co-network. The amount of time to use for these timers is taken from the PIN KEY TIMER INTERVAL and PIN KEY TIMER VALUE fields on KEYF screen 3.

If a Key Exchange timer expires, the BIC ANSI Interface performs as follows:

1. Determines the key that needs to be exchanged.
2. Changes the status of the co-network to indicate that a key exchange is pending.
3. Determines the type of message to send based on the value in the KEY PROCESSING TYPE field on KEYF screen 3 and the key direction, as described below:
 - a. If the KEY PROCESSING TYPE field contains the value N (no key processing), then no further steps are taken.
 - b. If the KEY PROCESSING TYPE field contains the value M (master processor), the message type is a new key message.
 - c. If the KEY PROCESSING TYPE field contains the value S (slave processor), the message type is a change key message.

- d. If the KEY PROCESSING TYPE field contains the value C (co-network processor) and the key direction is outbound, the message type is a new key message.
 - e. If the KEY PROCESSING TYPE field contains the value C (co-network processor) and the key direction is inbound, the message type is a new key message.
4. Determines whether a key exchange is already in progress for the specific key. If a key exchange is already in progress for the specific key, no further steps are taken. If a key exchange is not in progress for the key, processing continues with the next step (step 5).
5. Determines whether there is a station available to which to send the message. Depending on whether a station is available, the BIC ANSI Interface performs as follows:
 - a. If no stations are available, the BIC ANSI Interface sets an Extended Network Management Message timer. When this timer expires, the BIC ANSI Interface will attempt to send the message again. No further processing is performed at this time.
 - b. If a station is available, the BIC ANSI Interface performs according to the type of message, as described below:
 - 1) If the message is a new key message, the BIC ANSI Interface sends the message to the selected station. The BIC ANSI Interface sets a Network Management Message timer to wait for the response. The BIC ANSI Interface also sets a flag to indicate that a key exchange is in progress and deletes the Clear Old Key timer for the key being exchanged (if one is still set).
 - 2) If the message is a change key message, the BIC ANSI Interface sends the message to the selected station. The BIC ANSI Interface sets a Network Management Message timer to wait for the response.
6. Adds one to the count of messages outstanding to the station.

Note: Upon receiving a response from the co-network, the BIC ANSI Interface clears the flag that indicated a key exchange was pending and sets a new Key Exchange timer and Clear Old Key timer.

Clear Old Key Timer

The Clear Old Key timer limits the amount of time that the Authorization process attempts to use the old key to verify a PIN after a key exchange has taken place. The old key can be used in a second PIN verify attempt by the Authorization process or during a second attempt at translating the PIN that is outbound to a switch or another host.

The INBOUND PIN KEY and OUTBOUND PIN KEY fields on KEYF screen 2 can each contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key is currently in use. At the time of a successful key exchange, the CURRENT INDEX is updated to point to the new key and the key that was in use becomes the old key.

When processing a transaction, the BIC ANSI Interface uses the new key for PIN translation. If the security device cannot verify the PIN because of a key error, the PIN translation is tried again using the old key, if one exists. This timer controls how long the old key exists. When the timer expires, the old key is set to zeros.

A Clear Old Key timer is set under the following circumstances:

- When the BIC ANSI Interface receives a duplicate new key message from the co-network (carrying a key that was already received in a previous new key message and stored in the KEYF.)
- When the BIC ANSI Interface approves a new key message from the co-network.
- When the BIC ANSI Interface receives an approved new key response message from the co-network (i.e., BASE24 sent the co-network a new key, and the co-network has responded that the new key has been implemented successfully).

When a Clear Old Key timer expires, the BIC ANSI Interface performs as follows.

Note: KEYF information is read into memory by the BIC ANSI Interface at initialization, and the BIC ANSI Interface performs the following steps on the KEYF information in memory and in the KEYF file.

1. Determines the type of key for which the timer was set.
2. Determines the key to clear using the value in the CURRENT INDEX field.

Since the value in the CURRENT INDEX field points to the newest key, the BIC ANSI Interface selects the key not being pointed to by the CURRENT INDEX field as the one to clear.

3. Sets the selected key value and its check digits in memory to ASCII zeros.
4. Sets the selected key value and its check digits in the KEYF to binary zeros.

Report Startup Timer

The BIC ANSI Interface uses a Report Startup timer to start the SETL01 and STAT09 reports. The BIC ANSI Interface starts the timer at initialization and at interchange cutover. The time for this timer is taken from the LCONF param SW-RPT-STARTUP.

When the Report Startup timer expires, the obey file containing the commands to run the report programs is executed.

Configuring Key Management

The BIC ANSI Interface reads the Key File (KEYF) for the information it needs to handle PIN encryption, PIN translation, and dynamic key management. The KEYF contains one record for each interchange/BIC ANSI Interface combination. The ***BASE24 Transaction Security Manual*** describes in detail the configuration of PIN encryption, PIN translation, and dynamic key management. The key management options available for the BIC ANSI Interface are described briefly in the paragraphs that follow.

Interchange Encryption Type

The institution can specify the type of PIN encryption used for messages going to and coming from the co-network. There are three options:

- Clear PIN processing. All PINs handled by the co-network are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by the co-network are encrypted under the secure keys of a security module. The security modules that can be used include the Atalla Security Module (models MN10150-A1, MN10150-A2, and MN10150-A3), the Atalla Network Security Processor, and the Racal-Guardata Host Security Module (formerly VISA Security Module).
- Encrypted PINs—software DES PIN processing. All PINs handled by the co-network are encrypted via software, using the DES algorithm. No security module is in use.

This option is controlled via the ENCRYPT TYPE field on KEYF screen 1.

BASE24 Encryption Type

BASE24 and the co-network are not required to handle PIN encryption in the same manner. Each has its own PIN encryption options that are controlled separately. BASE24 can handle PIN encryption in two different ways:

- Clear PIN processing. All PINs handled by BASE24 are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by BASE24 are encrypted under the secure keys of a security module.

This option is controlled via the BASE24 ENCRYPT TYPE field on KEYF screen 1.

PIN Block Type

The institution can specify the type of PIN block used during encryptions. PINs are not simply fed into the encryption routine as is. Rather, they are presented to the encryption routines in what is called a *PIN block*. A PIN block is a 16-digit series of hexadecimal characters, of which the PIN is a part.

The BIC ANSI Interface supports two types of PIN blocks—ANSI PIN blocks and PIN/PAD PIN blocks. The interchange can specify one of these two PIN block types or specify clear PINS, indicating PINs are not encrypted.

This option is controlled via the PIN BLOCK FORMAT field on KEYF screen 1.

PAN Characters Used in PIN Block

If the ANSI PIN block is selected, the institution can specify which PAN characters to use in formatting the PIN block. There are three options:

- Use the 12 right-most digits of the PAN, excluding the check digit
- Use the 12 right-most digits of the PAN, including the check digit
- Use the 12 left-most digits of the PAN

This option is controlled via the ANSI PAN FORMAT field on KEYF screen 1.

PAD Character Used in PIN Block

If the PIN/PAD PIN block is selected, the institution must specify the PAD character to use in formatting the PIN block. The PAD character is a single hexadecimal character used to fill out the remaining positions of the PIN block.

The PAD character is specified via the PIN PAD CHARACTER field on KEYF screen 1.

Number of Keys

The number of keys indicates whether the inbound and outbound keys supported by the BIC ANSI Interface process are the same or different. If the keys are the same, only 1 key is generated and used for both inbound and outbound keys. If the keys are different, a separate key is generated for inbound and outbound keys. This field is used to determine the number of keys being used for PINs and message authentication.

The number of different PIN keys supported by the BIC ANSI Interface process is specified via the NUMBER OF KEYS field on KEYF screen 1.

Key Length

The key length indicates the length of the Key Encryption Keys (KEKs) supported by the BIC ANSI Interface process. This field identifies whether the BIC ANSI Interface process supports single-length KEKs or double-length KEKs. Single-length KEKs consist of one 64-bit key. Double-length KEKs consist of a key pair consisting of 64 bits each. For more information on single-length and double-length KEKs, refer to the ***BASE24 Transaction Security Manual***.

The type of key management supported by the BIC ANSI Interface process is specified via the KEY LENGTH field on KEYF screen 1.

PIN Key Variant

The PIN key variant indicates whether to translate the PIN key directly from the PIN Key Exchange Key (KEK) parts or to include the variant 1 constant with the PIN KEK parts.

The PIN key variant option supported by the BIC ANSI Interface process is specified via the PIN KEY VARIANT field on KEYF screen 3.

Key Processing Type

The key processing type describes whether the BIC ANSI Interface process is performing master, slave, co-network, or no key processing. The following list describes each key processing type:

- Master — Indicates the process as controlling all key generation.
- Co-network — Indicates that this process shares the control of key management processing with its co-network. Each co-network is only responsible for exchanging its outbound key.
- Slave — Indicates that this process is not responsible for any key generation, although it can request that a key be generated.
- None — No key processing is supported.

The key processing type supported by the BIC ANSI Interface process is specified via the KEY PROCESSING TYPE field on KEYF screen 3.

PIN Management Keys

There are four types of keys identified by BASE24 for use in managing PINs between the BIC ANSI Interface and an interchange: inbound encryption keys, outbound encryption keys, intermediate keys, and exchange keys. The institution must specify the values for these keys.

These options are controlled via KEYF screen 2.

Cutover and Reconciliation Handling

The following options control how the BIC ANSI Interface handles cutover and reconciliation with the co-network.

Reconciliation Totals

The BIC ANSI Interface can be set up to specify whether it should maintain reconciliation totals to be used during cutover processing. This option is controlled via the PROCESSING MODE field on ICF screen 3.

Cutover Relationship

The cutover relationship between BASE24 and the co-network can be defined as an equal partner or a master/slave relationship. This option is controlled via the CUTOVER STATUS field on ICF screen 12.

Not-On-Us Transactions

The BIC ANSI Interface can be set up to specify the conditions under which certain incoming transactions are allowed. Transactions initiated on the co-network can be limited by transaction type. Individual transaction types can be disallowed to the co-network, or limited to being accepted only within the same county, state, or nation. These options are controlled via the NOT ON US fields on ICF screens 7 and 9.

Transactions Allowed

The BIC ANSI Interface can be set up to specify the transactions that can be sent to the co-network for authorization. An individual transaction type is either allowed or not allowed to the co-network. These options are controlled via the TRANSACTION ALLOWED TO SWITCH fields on ICF screens 7 and 9.

Stand-in Authorization

The BIC ANSI Interface can be set up to specify whether the BASE24 can stand in to authorize transactions if the link to the authorizing entity is down. This option is controlled via the STAND-IN AUTH field on ICF screen 12.

Customer Balance Display

The BIC ANSI Interface can be set up to specify whether customer balance information for incoming responses from the co-network should be displayed, printed, or both displayed and printed. This option is controlled via the CUSTOMER BALANCE DISPLAY field on ICF screen 1. It is only used when a transaction is authorized by the co-network.

Automatic Logon

The BIC ANSI Interface can be set up to automatically log on to the co-network at startup. If this option is selected, no operator intervention is required to log on to the co-network. It is done automatically as part of the initialization of the BIC ANSI Interface process. This option is controlled via the AUTO SIGNON START field on ICF screen 3. If automatic logon is used, and dynamic key management is supported in conjunction with a security device, PIN keys are automatically exchanged at logon.

Product Indicators

The BIC ANSI Interface can be set up to specify which products it can support. The BIC ANSI Interface currently supports BASE24-atm and BASE24-pos. This option is controlled via the PRODUCT INDICATOR fields on ICF screen 12.

Defining Stations

The BIC ANSI Interface can be connected to a maximum of 2 stations. These stations are defined via the STATION 1 and STATION 2 fields on ICF screen 1. The stations defined in the ICF must also have a corresponding declaration in the Network Environment Source File (NEFS).

ICF Screens

This section contains a listing of ICF fields used by the BIC ANSI Interface. ICF screens 1 through 11 are shared by multiple Interchange Interfaces. ICF screen 12 is interchange-specific, which means there is an ICF screen 12 used exclusively for the BIC ANSI Interface.

The following tables indicate the ICF fields used by the BIC ANSI Interface and the expected value. Values in quotation marks (“ ”) are literal values that must be placed in the corresponding fields. The BIC ANSI Interface does not use all the fields on all the ICF screens. These fields are noted with “not used” in the Values column.

For documentation on ICF screens 1 through 11 and their corresponding field descriptions, refer to the ***BASE24 Base Files Maintenance Manual***. ICF screen 12 and descriptions of its fields are provided later in this section.

Note: The information contained in the following tables does **not** indicate whether a field is required, but rather if a field can be used if you so choose. The field may or may not be required.

ICF Screen 1

ICF screen 1 contains fields establishing interchange identifiers, station names, default values for certain fields, and report title names to be printed on the SETL01-BIC and STAT09-BIC reports. The fields on ICF screen 1 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 1	
Field	Value
INTERCHANGE FIID	Interchange FIID
PROCESS	Process name as defined in the NEFS
SWITCH TYPE	“BIC”
INTERCHANGE LOGICAL NET	Logical network identifier
REPORTING NAME	Not used

ICF SCREEN 1	
Field	Value
INSTITUTION ID	Not used
SWITCH ID	Not used
STATION 1	Symbolic station name
STATION 2	Symbolic station name
SIC CODE	SIC code identifying the co-network
CURRENCY CODE	“840”
DEFAULT TERM NUM	Not used
DEFAULT ACQUIRER ID NUM	Not used
CUSTOMER BALANCE DISPLAY	0, 1, 2, or 3

ICF Screen 2

ICF screen 2 contains interchange settlement information and pertinent holiday dates. The fields on ICF screen 2 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 2	
Field	Value
SETTLEMENT HOUR	00–23
SETTLEMENT MINUTE	00–59
SETTLEMENT DAYS	“0”
REPORT PRIORITY	0–255
REPORT CPU	CPU number
SWITCH POSTING DATE	YYMMDD
HOLIDAY DATES	YYMMDD

ICF Screen 3

ICF screen 3 is split up into two sections. The first section defines transaction message timer limits used by the BIC ANSI Interface. The second section defines the processing options used by the BIC ANSI Interface. The fields on ICF screen 3 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 3	
Field	Value
NETWORK MANAGEMENT	0–9999 (seconds)
ACQUIRER	Y or N
EXTENDED NETWORK	0–9999 (seconds)
ISSUER	Y or N

ICF SCREEN 3	
Field	Value
WAIT FOR TRAFFIC	0–9999 (seconds)
PROCESSING MODE	1. (If this field is set to 0, BASE24 will not keep totals. ACI recommends this field be set to 1.)
PERFORMANCE PERIOD	0–9999 (minutes in performance monitoring period)
AUTO SIGNON START	Y or N
MAXIMUM TIMEOUTS	0–9999 (timeouts allowed)
MAX SAF RETRY	0–9999 (SAF retries allowed)
OUTBOUND	0–9999 (outbound transactions allowed)
ACK TO SWITCH	Y or N
INBOUND	0–9999 (inbound transactions allowed)
ACK FROM SWITCH	Y or N
NETWORK MANAGEMENT MESSAGE ENABLED	Y or N
TYPE OF INTERCHANGE REPORTS	Not used
ILF EXTRACT NUMBER	Not used

ICF Screen 7

ICF screen 7 defines ATM transaction sharing parameters and ATM transactions allowed to the co-network. To use ICF screen 7, the ATM flag on ICF screen 12 must be set to Y. The fields on ICF screen 7 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 7	
Field	Value
NOT ON US	0, 1,2, 3, or 4
TRANSACTIONS ALLOWED TO SWITCH	Y or N

ICF Screen 8

ICF screen 8 defines ATM transaction message timer limits, ATM Authorization process name, and ATM sharing groups. The fields on ICF screen 8 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 8	
Field	Value
AUTH PROCESS	Authorization process name
DEFAULT ROUTING GROUP	Terminal routing group number
STORE AND FORWARD	0–9999 (seconds)
OUTBOUND	0–9999 (seconds)
INBOUND	0–9999 (seconds)
COMPLETION	0–9999 (seconds)
COMPLETION ACK	Not used
SHARING GROUPS	Sharing group codes

ICF Screen 9

ICF screen 9 defines POS transactions allowed to the co-network. The fields on ICF screen 9 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 9	
Field	Value
TRANSACTIONS ALLOWED TO SWITCH	Y or N

ICF Screen 10

ICF screen 10 defines POS transaction message timer limits, POS Authorization process name, POS default retailer ID, and POS timeout action. The fields on ICF screen 10 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 10	
Field	Value
AUTH PROCESS	Authorization process name
REFERRAL PHONE NUMBER	Telephone number
RETAILER ID DEFAULT	Default retailer ID
TIMEOUT ACTION	Not used
SETTLE ENTITY	Not used
STORE AND FORWARD	0–9999 (seconds)
OUTBOUND	0–9999 (seconds)
INBOUND	0–9999 (seconds)
COMPLETION	0–9999 (seconds)
COMPLETION ACK	Not used

ICF Screen 11

ICF screen 11 defines POS preauthorization default time and amount, approval code length, and services allowed to the co-network. The fields on ICF screen 11 used by the BIC ANSI Interface and their expected values are indicated in the following table.

ICF SCREEN 11	
Field	Value
DEFAULT PRE-AUTH AMOUNT	Not used
APPROVAL CODE LENGTH	2–6
PRE-AUTH HOLD INCREMENT	Not used
PRE-AUTH HOLD TIME	00–99
ALLOWED SERVICES	Allowed services code

ICF Screen 12

ICF screen 12 is an interchange-specific screen used exclusively by the BIC ANSI Interface. It defines options used at logon to determine processing requirements for the BIC ANSI Interface. All fields on ICF screen 12 can be used by the BIC ANSI Interface. ICF screen 12 is shown below followed by descriptions of its fields.

```

BASE24-INFC ICF FILE MAINTENANCE      LLLL      YY/MM/DD  HH:MM  12 OF 13
INTERCHANGE FIID:                      PROCESS:

                                     BASE24

      CO-NETWORK RELEASE: 1

      COMPLETIONS: 0

      COMPLETION ACKS: 0

      STAND-IN AUTH: 0

      REVERSAL ACK SUPPRESS: 0

      REVERSAL AFTER COMP SUPPRESS: 0

      CUTOVER STATUS: 0
      PRODUCT INDICATOR (Y/N): N (ATM)  N (POS)

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                                     F12-HELP

```

CO-NETWORK RELEASE — A code identifying the release of the BIC ANSI Interface process used by the co-network. This code enables this BIC ANSI Interface process to convert the previous release response codes used by the co-network into current release response codes. Valid values are as follows:

- 0 = Previous release
- 1 = Current release
- 2 = Current release with address verification

Note: BASE24 does not distinguish between CO-NETWORK RELEASE code values of 1 and 2 because address verification processing has no impact on response codes.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: ICFBIC.CO-NET-REL

COMPLETIONS — A code specifying whether completions, or text-level acknowledgments, are required after an approved timely response to an authorization request. If either BASE24 or the co-network require completions, both must require completions. When completion messages are required, they are sent to the issuer network by the acquirer network. Valid values are as follows:

- 0 = Completions are not required
- 1 = Completions are required

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: ICFBIC.COMPLETIONS

COMPLETION ACKS — A code specifying whether completion acknowledgments are required in response to a completion message. If either BASE24 or the co-network requires completion acknowledgments, both must require completion acknowledgments. When completion acknowledgment messages are required, they are sent to the acquirer network by the issuer network. Valid values are as follows:

- 0 = Completion acknowledgments are not required
- 1 = Completion acknowledgments are required

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: ICFBIC.COMPLETION-ACKS

STAND-IN AUTH — A code specifying whether the acquirer network can authorize transactions for the issuer network when the communications link with the issuer network is down. When stand-in authorization is allowed, a timed out or link down transaction is returned to the acquiring BASE24 Authorization process as a failed 0200 message. When stand-in authorization is not allowed, a timed out or link down transaction is denied. When this option is selected, a force post message is forwarded to the issuer network when communications are

reestablished. If completion acknowledgment messages are required, the issuer network must send a completion acknowledgment message in response to the force post message. Valid values are as follows:

- 0 = Do not perform stand-in authorization
- 1 = Perform stand-in authorization

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: ICFBIC.STAND-IN

REVERSAL ACK SUPPRESS — A flag specifying whether to suppress text-level acknowledgments to reversal messages. If either BASE24 or the co-network suppresses reversal acknowledgments, both must suppress reversal acknowledgments. When this option is chosen, the issuer network does not send a text-level acknowledgment to the acquirer network in response to a reversal message. Valid values are as follows:

- 0 = Do not suppress reversal acknowledgments
- 1 = Suppress reversal acknowledgments

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: ICFBIC.RVSL-ACK-SUPPRESS

REVERSAL AFTER COMP SUPPRESS — A code specifying whether to suppress text-level reversal messages after a completion message has been sent. This option is only possible if completion messages are required. If either BASE24 or the co-network suppresses reversal messages after completion messages, both must suppress reversals after completions. When this option is chosen, reversals are logged to the ILF (or similar file on the co-network side) by the acquirer network but are not sent to the issuer network. The reversals must be manually posted by the issuer network to the customer accounts. Valid values are as follows:

- 0 = Do not suppress reversals after completions
- 1 = Suppress reversals after completions

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: ICFBIC.RVSL-SUPPRESS

CUTOVER STATUS — A code specifying the equality of the co-network with regard to cutover and settlement. BASE24 and the co-network can be equal partners or one network can be the master network and the other network the slave network.

If BASE24 and the co-network are equal partners regarding cutover and settlement, the BIC ANSI Interface sends settlement totals to its counterpart when its own BASE24 network cuts over. The BIC ANSI Interface expects to receive cutover notification and settlement totals from the co-network when it cuts over.

If BASE24 and the co-network are not equal partners regarding cutover and settlement, this field indicates whether this network is the master or the slave. The master network sends the slave network cutover notification and settlement totals when it cuts over. In this case, the slave network is not responsible for sending cutover notification and settlement totals. Valid values are as follows:

0 = Equal
1 = Master
2 = Slave

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: ICFBIC.CTVR-STAT

PRODUCT INDICATOR (Y/N)

The following flags identify which BASE24 product messages this interchange can process.

ATM — A flag identifying whether BASE24-atm messages are processed by this interchange. Valid values are as follows:

Y = Yes, BASE24-atm messages are processed
N = No, BASE24-atm messages are not processed

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: ICFBIC.PROD-IND

POS — A flag identifying whether BASE24-pos messages are processed by this interchange. Valid values are as follows:

Y = Yes, BASE24-pos messages are processed
N = No, BASE24-pos messages are not processed

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: ICFBIC.PROD-IND

KEYF Screens

This section contains a listing of KEYF fields used by the BIC ANSI Interface.

The following tables indicate the KEYF fields used by the BIC ANSI Interface and the expected value. Values in quotation marks (“ ”) are literal values that must be placed in the corresponding fields. The BIC ANSI Interface does not use all the fields on all the KEYF screens. These fields are noted with “Not used” in the Values column.

For documentation on the KEYF screens and their corresponding field descriptions, refer to the ***BASE24 Files Maintenance Manual***. For information on BASE24 security, refer to the ***BASE24 Transaction Security Manual***.

Note: The information contained in the following tables does **not** indicate whether a field is required, but rather if a field can be used if you so choose. The field may or may not be required.

KEYF Screen 1

KEYF screen 1 defines the number of PIN keys currently being used by the interface, the length of the Key Exchange Keys (KEKs) used by the interface, and the PIN exchange keys used by the interface. In addition, screen 1 contains security information used to set up the PIN parameters associated with the BIC ANSI Interface process. The fields on KEYF screen 1 used by the BIC ANSI Interface and their expected values are indicated in the following table.

KEYF SCREEN 1	
Field	Value
DPC/FIID	Financial institution identifier
INTERFACE PROCESS	Interface process name
ENCRYPT TYPE	0 for clear PINs, 1 for Security module PIN management, 2 for software DES PIN management. This field must contain a 1 if using dynamic key management.
BASE24 ENCRYPT TYPE	0 for clear PINs, 1 for Security module PIN management

KEYF SCREEN 1	
Field	Value
MAC ENCRYPT TYPE	Not used
ANSI PAN FORMAT	0 for the last 12 digits of the PAN, excluding the check digit; 1 for the last 12 digits of the PAN, including the check digit; or 2 for the first 12 digits of the PAN.
PIN BLOCK FORMAT	0 for clear PINs, 1 for ANSI (PIN/PAD) PIN block, 3 for PIN/PAD PIN block. This field must contain a 1 or 3 if using dynamic key management.
PIN PAD CHARACTER	A–F
MAC DATA TYPE	Not used
NUMBER OF KEYS	1 if the same key is used for inbound and outbound messages, or 2 if different keys are used for inbound and outbound messages or if the KEY PROCESSING TYPE field on KEYF screen 3 is set to C.
KEY LENGTH	1
FULL MESSAGE MAC	Not used
INTERMEDIATE KEYS: CLEAR	Clear version of Intermediate key
INTERMEDIATE KEYS: ENCRYPTED	Encrypted version of Intermediate key
INTERMEDIATE KEYS: CHECK DIGITS	0–9, A–Z
EXCHANGE KEYS: PIN KEY	Two sets of 16 hexadecimal characters
EXCHANGE KEYS: CHECK DIGITS	4 hexadecimal characters

KEYF SCREEN 1	
Field	Value
EXCHANGE KEYS: MAC KEY	Not used
EXCHANGE KEYS: CHECK DIGITS	Not used

KEYF Screen 2

KEYF screen 2 contains inbound and outbound working key information for PINs. The fields on KEYF screen 2 used by the BIC ANSI Interface and their expected values are indicated in the following table.

KEYF SCREEN 2	
Field	Value
OUTBOUND KEYS: PIN KEY1	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
OUTBOUND KEYS: CURRENT INDEX 1	1 for the value in the PIN KEY1 field, or 2 for the value in the PIN KEY2 field.
OUTBOUND KEYS: PIN KEY2	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
OUTBOUND KEYS: KEY COUNTER	6 hexadecimal characters
INBOUND KEYS: PIN KEY1	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
INBOUND KEYS: CURRENT INDEX 1	1
INBOUND KEYS: PIN KEY2	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
INBOUND KEYS: KEY COUNTER	6 hexadecimal characters

KEYF SCREEN 2	
Field	Value
OUTBOUND KEYS: MAC KEY1	Not used
OUTBOUND KEYS: CHECK DIGITS1	Not used
OUTBOUND KEYS: CURRENT INDEX 1	1
OUTBOUND KEYS: MAC KEY2	Not used
OUTBOUND KEYS: CHECK DIGITS2	Not used
OUTBOUND KEYS: KEY COUNTER	Not used
INBOUND KEYS: MAC KEY1	Not used
INBOUND KEYS: CHECK DIGITS1	Not used
INBOUND KEYS: CURRENT INDEX 1	1
INBOUND KEYS: MAC KEY2	Not used
INBOUND KEYS: CHECK DIGITS2	Not used
INBOUND KEYS: KEY COUNTER	Not used

KEYF Screen 3

KEYF screen 3 contains information on the timer, threshold, and processing parameters used by the interface for Dynamic Key Management (DKM). It also identifies the originator and receiver of DKM messages. The fields on KEYF screen 3 used by the BIC ANSI Interface and their expected values are indicated in the following table.

Note: If all limits on this screen are set to zero, the BIC ANSI Interface process using this KEYF record will not initiate key changes automatically based on timers or counters, so the co-network must be responsible for any key changes if dynamic

key management is to occur. At least one PIN key limit on this screen must be set to a nonzero value for the BIC ANSI Interface to initiate dynamic key management for PINs.

KEYF SCREEN 3	
Field	Value
PIN KEY VARIANT	0 for PIN translation directly from the PIN KEK parts, or 1 for translation that includes the variant 1 constant with the PIN KEK parts.
MAC KEY VARIANT	Not used
PIN KEY TIMER VALUE	0 if not used; 5–1500 if timer interval is minutes; 0–1000 if timer interval is hours; or 0–180 if timer interval is days.
MAC KEY TIMER VALUE	Not used
PIN KEY TIMER INTERVAL	D for days, H for hours, or M for minutes.
MAC KEY TIMER INTERVAL	Not used
PIN KEY TRAN	0 if a total transaction limit is not used or 50–100000.
MAC KEY TRAN	Not used
PIN KEY ERROR	0 if an error limit is not used or 5–9999
MAC KEY ERROR	Not used
CONSECUTIVE PIN KEY ERROR	0 if a consecutive error limit is not used or 5–9999
CONSECUTIVE MAC KEY ERROR	Not used
MAC SYNCHRONIZATION ERROR	Not used
CLEAR OLD KEY TIMER VALUE	0 if the old key is not cleared or 1–9999 (seconds)

KEYF SCREEN 3	
Field	Value
ORIGINATING ID	ID of sender of dynamic key management messages
RECEIVING ID	ID of receiver of dynamic key management messages
KEY PROCESSING TYPE	C if each interface is only responsible for generating its own outbound keys, M if the interface is responsible for all key generation, S if the interface receives keys but does not generate them, or N if dynamic key management is not used.
NOTARIZATION SUPPORTED	Not used

NETGEN Guidelines

The BIC ANSI Interface requires a minimum of one process, station, device, and line declaration each in the NEFS. Sample declarations are provided below.

```
PROCESS P1A^BIC:                                ! BIC SWITCH PROCESS
    OBJECT "$DATA.SW500BJ.BIC",                  !
    LIBRARY "$SYSTEM.SPANNET.SKELB",             !
    PPD NAME IS "$T1AB",                         !
    LOGICAL NETWORK IS BIC,                      !
    SYMBOLIC USAGE,                             !
    BOJ,                                          !
    MAX CORE QUEUE 100,                          !
    PRIORITY 139,                                !
    CPU 0 & 1;                                   !

DEVICE BIC^SWITCH = 29:                          ! "BIC" SWITCH DEVICE
    RECEIVE 1000,                                !
    TRANSMIT 4000,                                !
    TRANSMISSION NUMBER LENGTH 0,                !
    NORMAL SELECT;                               !

STATION S1ABIC = 05:                             ! STATION TO BE USED FOR
    DEVICE BIC^SWITCH,                           !     BIC SWITCH
    LOGICAL NETWORK IS BIC,                      !
    DESTINATION PROCESS P1A^BIC,                 !
    ADDRESS "BB":"SB",                           ! FOR IBM SERIES 1 INTERFACE
    MAX CORE QUEUE 3,                            !
    TERMINAL TYPE 0,                             !
    RETRY 3,                                     !
    REINSTATE AFTER 60 SECONDS,                  !
    NO LOGICAL ACK;                             !

LINE L1A^BIC1 = 05:                              !
    DEVICE ID "$BSCD1",                          !
    BI-SYNC POINT-TO-POINT PRIMARY,              !
    SYNC,                                         !
    TIMEOUT 3 SECONDS,                          !
    RETRY 1,                                     !
    POLL INTERVAL 1 SECOND,                     !
    OUTPUT BIAS 4,                               !
    ERROR ANALYSIS INTERVAL 60 SECONDS,         !
    STATIONS (                                   !
        S1ABIC );                               !
```

For a complete descriptions of NEFS declarations and all NETDL elements, refer to the ***BASE24 Network Environment Source File Manual***.

Section 3

Network Management and Operations

Network management messages can be divided into two categories: those used to manage the operational status of the communications lines between BASE24 and the co-network, and those used to manage PIN and message authentication keys between BASE24 and the co-network.

The BIC ANSI Interface supports four types of network management messages:

- Logon messages
- Echo test messages
- Logoff messages
- Dynamic key management (DKM) messages

Network management messages are sent as 0800 messages and require 0800 messages in response.

This section describes the message flows that occur when network management messages are sent.

Monitoring Co-Networks and Stations

The BIC ANSI Interface process keeps track of the status of all co-networks and stations with which it communicates via its station table. This station table is built at initialization and is updated continuously during processing.

Stations are marked *up* and *down* in the table depending on whether the BIC ANSI Interface process can communicate with them. For instance, if a station fails to respond appropriately to a message, the BIC ANSI Interface process checks the maximum retries allowed for the co-network's stations. If the maximum retries allowed is reached, the BIC ANSI Interface process marks the station down and takes the necessary network management actions to reestablish the link with the station. Once the link is reestablished, the BIC ANSI Interface process marks the station up.

The BIC ANSI Interface process also tracks the status of the co-networks. A co-network is marked as *up* if the BIC ANSI Interface process can send messages to and receive messages from at least one of the co-network's stations. If all of the stations to a co-network are marked down, the co-network itself is marked down. Additionally, the BIC ANSI Interface process tracks such things as whether the co-network is logged off, whether store-and-forward processing is in progress for the co-network, and whether a SAF extract is in progress for the co-network.

The BIC ANSI Interface process keeps track of station and co-network information at all times and logs a message when the status of the co-network changes (for example, from up to down or at the start of store-and-forward processing).

Note: The BIC ANSI Interface process only tracks the logical ability of BASE24 to communicate with each station. If a station responds to a message, the BIC ANSI Interface process considers it to be up. If a station fails to respond to a message the number of times specified in the MAXIMUM TIMEOUTS field on ICF screen 3, the BIC ANSI Interface process considers the station to be down.

Also, echo test messages do not substitute for logon messages. The BIC ANSI Interface process does not consider the co-network to be logged on and available for processing transactions until it receives a logon message.

Logon Messages

Logon messages are network management messages used to manage communications access between BASE24 and a co-network. They are sent as 0800 messages and require 0800 messages in response.

Logon messages are distinguished by a value of 001 in the Network Management Information Code field of the network management message.

Logon Messages from BASE24

The BIC ANSI Interface sends logon messages to the co-network in return for echo test messages. This is done regardless of whether the co-network initiated the echo test message or was simply responding to a prior BASE24-initiated echo test message with an echo test message.

Logon Messages from a Co-Network

Co-networks send logon messages to start a log on sequence after a period of being logged off. They can also send logon messages in response to BASE24-initiated echo tests or logon messages.

When the BIC ANSI Interface process receives a logon message after the co-network has been logged off for a period, the BIC ANSI Interface process marks the co-network as logged on and begins sending echo test messages to its stations.

Note: If the BIC ANSI Interface receives a logon message when the co-network is not logged off, it simply drops the message.

When the BIC ANSI Interface receives a logon message in response to a BASE24-initiated message, the BIC ANSI Interface marks the station up and begins processing.

Echo Test Messages

Echo test messages are network management messages used to establish whether a station is available for message processing. They are sent as 0800 messages and require 0800 messages in response.

Echo tests messages are distinguished by a value of 301 in the Network Management Information Code field of the network management message.

Echo Test Messages from BASE24

The BIC ANSI Interface process initiates echo tests under the following circumstances:

1. At initialization. The BIC ANSI Interface sends an echo test to each send/receive station at initialization.
2. Upon receipt of a Logon command from the Network Control Supervisor (NCS) screen. The BIC ANSI Interface sends an echo test to each station associated with the co-network included in the command.
3. Upon expiration of an Extended Network Management Message timer. This processing is described in section 2.
4. Upon expiration of a Wait-for-Traffic timer. This processing is described in section 2.
5. When a co-network initiates a logon after a period of being logged off. The BIC ANSI Interface process sends an echo test to each send/receive station once the co-network has been marked as logged on.

Echo Test Messages from a Co-Network

Co-networks can send echo test messages when they wish to start a logon sequence after a period of being logged off. Or, they may send echo test messages in response to BASE24-initiated echo test messages.

Upon receipt of an echo test message from a co-network, the BIC ANSI Interface always returns a logon message.

Logoff Messages

Logoff messages are network management messages used to stop communications between BASE24 and a co-network. They are sent as 0800 messages and do not require a response.

Logoff messages are distinguished by a value of 002 in the Network Management Information Code field in the network management message. They can be sent by either BASE24 or a co-network.

In either situation, once a co-network is logged off, BASE24 does not send echo test messages or non-network management messages to any of the co-network's stations. Any non-network management messages that must be forwarded to the co-network are placed in the SAF instead.

BASE24 only attempts to reopen the link to a co-network that is logged off if a logon command is received from the Network Control SupervANSIr (NCS) screen. The co-network, on the other hand, can reopen the link at any time by sending an echo test or a logon message.

Note: When a co-network is logged off, any transactions in progress are allowed to complete.

Logoff Messages from BASE24

BASE24 only sends logoff messages to a co-network as a result of a logoff command issued from the NCS screen. BASE24 then expects a logoff response from the co-network.

To log a co-network off, BASE24 simply sends a logoff message to an available send/receive station and marks the co-network as down and logged off. A response may be returned by the co-network, but is not necessary.

Logoff Messages from a Co-Network

When a co-network must log off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations. Upon receipt of a logoff message, the BIC ANSI Interface process marks the co-network down and logged off. It does not return a response to the co-network.

Dynamic Key Management Messages

Dynamic key management messages make up a subset of network management (0800) messages. There are four key management messages available:

- Change key
- New key
- Repeat key
- Verify key

The change key and new key messages can be generated automatically as the result of a defined threshold being surpassed. In addition, all of these messages can be generated manually by an operator issuing a command from the Network Control Supervisor (NCS) screen.

This section provides an overview of how dynamic key management messages are used. For more information on dynamic key management, refer to the **BASE24 Transaction Security Manual**. For more information on using the NCS to generate key management messages, and for complete syntax of the NCS commands involved, see the **BASE24 Text Command Reference Manual**.

Change Key Message

A change key message requests that a key processor generate a new PIN encryption key. The receiving processor sends a response to the change key message and then, in a separate action, generates the new key and sends it to the requesting process in a new key message.

The BIC ANSI Interface sends this type of message to a co-network when the co-network is responsible for generating a particular key and the BIC ANSI Interface process has determined that that key needs to be changed. The BIC ANSI Interface could make that determination based on a DKM threshold being passed or in response to an NCS KEY CHANGE command (the NCS KEY CHANGE command has the same effect as passing a DKM threshold).

The BIC ANSI Interface process can receive this type of message from a co-network when the BIC ANSI Interface is responsible for generating a particular key and the co-network wants that key to be changed.

New Key Message

A new key message carries a new PIN encryption key between processors. The receiving processor verifies and begins using the new key and returns a response to the key-generating processor indicating that the new key has been implemented.

The BIC ANSI Interface sends this type of message to a co-network when the BIC ANSI Interface is responsible for generating a particular key and has either determined that that key needs to be changed or been requested by the co-network to change or repeat the key. This could occur as a result of a DKM threshold being passed, notification from the co-network (via a change key or repeat key message), or could occur in response to an NCS KEY CHANGE command (the NCS KEY CHANGE command has the same effect as passing a DKM threshold).

The BIC ANSI Interface process can receive this type of message from a co-network when the co-network is responsible for generating a particular key and the co-network has either determined that that key needs to be changed or been requested by the BIC ANSI Interface to change or repeat the key (via a change key or repeat key message).

Repeat Key Message

A repeat key message requests that a processor that generated a key resend that key. The processor receiving the repeat key message obtains the requested key from its database and sends it to the requesting process in a new key message. The key itself is not changed, nor are any threshold counters affected.

The BIC ANSI Interface only generates a repeat key message as the result of receiving an NCS KEY REPEAT command. It will respond to a repeat key message from the co-network with a new key message carrying the requested key.

Verify Key Message

A verify key message requests that a processor that generated a key verify the key by comparing check digits. The verify key message contains check digits for the key in question, and the processor receiving the message compares the check digits in the message with the check digits it has on file for the particular key. The processor then responds to the verify key message with the results of the check digit comparison. (If the check digits do not match, the processor that sent the verify key message should use a change key message to request a new key.)

The BIC ANSI Interface only generates the verify key message as the result of receiving an NCS KEY VERIFY command. It will respond to a verify key message from a co-network, by comparing check digits for the specified key and returning the results.

Interactive Network Management Message Flows

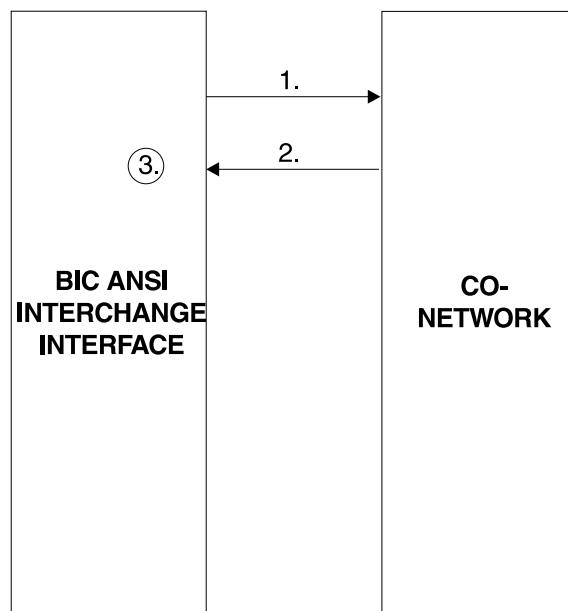
The remainder of this section contains examples of network management and dynamic key management message flows, and describes the processing performed by BASE24 for each.

Echo Test/Logon Message Examples

The following examples show echo test/logon sequences supported by BASE24.

Example #1

In this example, the BIC ANSI Interface is echo testing each of the co-network's send/receive stations (as it does at initialization), and the co-network is set up to respond to echo test messages with logon messages.

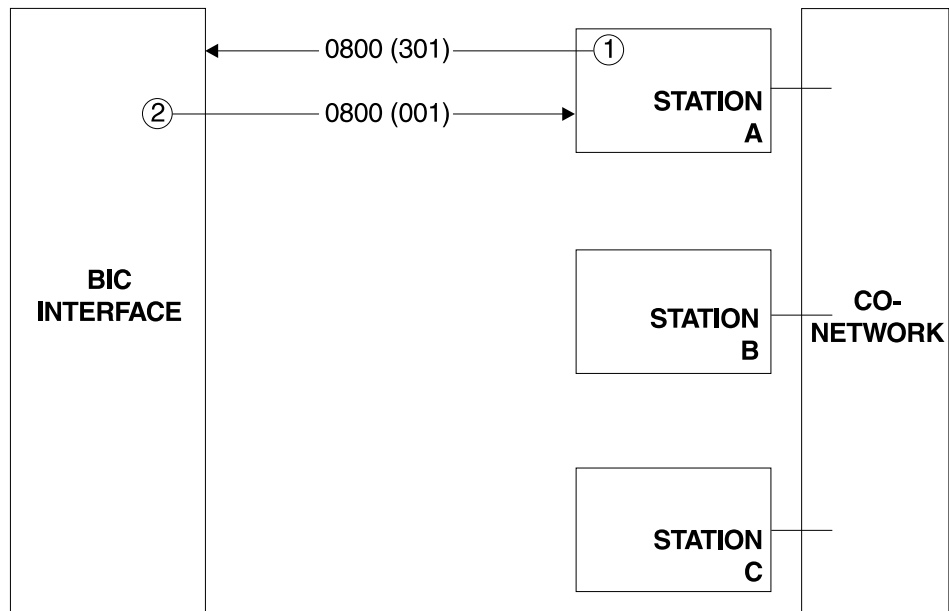


1. The BIC ANSI Interface sends an echo test message to each of the co-network's send/receive stations and sets a Network Management Message timer for each.
2. The co-network returns a logon message as a response from each station.

3. The BIC ANSI Interface receives each logon message, marks the corresponding station up, and sets a Wait-for-Traffic timer for the station
Upon receipt of the first logon message from the co-network's station, the BIC ANSI Interface also marks the co-network as up and logged on.

Example #2

In this example, Station A is the only station down, and the co-network sends an echo test.



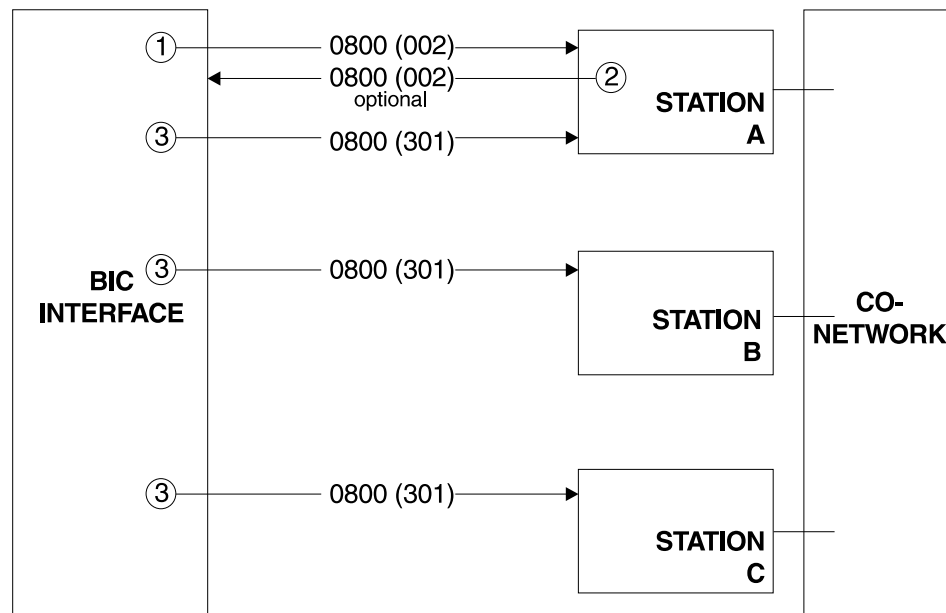
1. The co-network sends an echo test message from Station A.
2. The BIC ANSI Interface marks the station up, sets a Wait-for-Traffic timer for that station, and returns a logon message in response.

Logoff Message Examples

The following are two examples of logoff sequences supported by BASE24.

Example #1

In this example, the logoff and subsequent logon are both initiated by the BIC ANSI Interface.

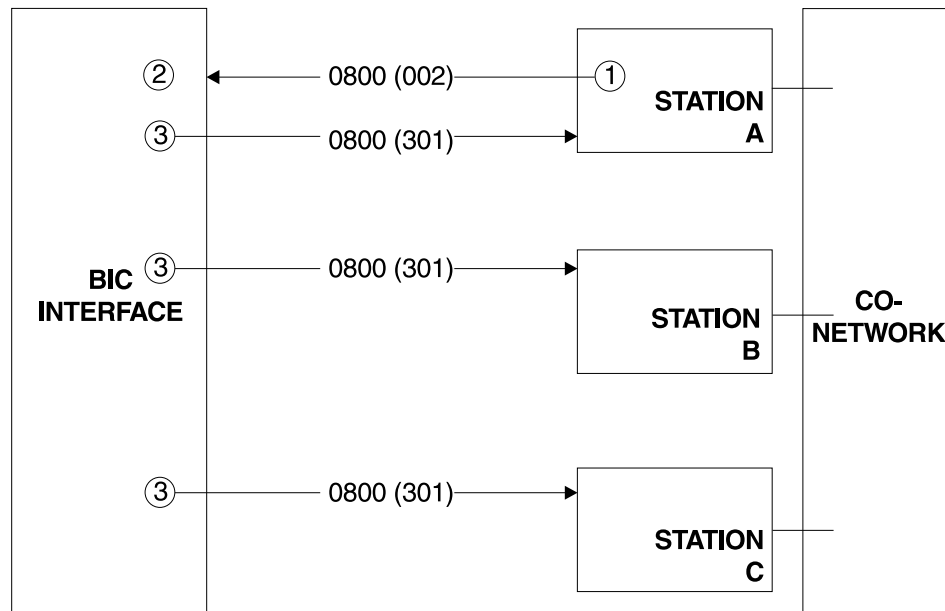


1. The BIC ANSI Interface receives a logoff command from the Network Control Supervisor (NCS) screen and performs the following steps:
 - a. Sends a logoff message to one of the co-network's stations.
 - b. Clears the following timers running for the co-network and its stations:
 - Network Management Message timers
 - Extended Network Management Message timers
 - Wait-for-Traffic timers
 - Performance timer
 - Key Exchange timers
 - Clear Old Key timersAll other timers are left running to ensure that current transactions in progress complete.
 - c. Marks all of the co-network's stations down.

- d. Marks the co-network as down and logged off.
2. The co-network may return a logoff message as a response, but it is not required.
3. The BIC ANSI Interface receives a logon command from the NCS and generates an echo test message to each of the co-network's send/receive stations.

Example #2

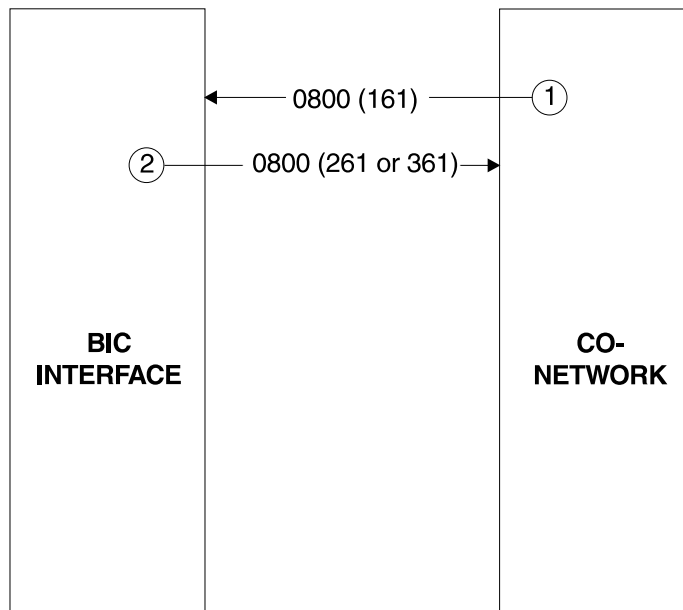
In this example, the logoff and subsequent logon are both initiated by the co-network:



1. The co-network sends a logoff message to BASE24.
2. The BIC ANSI Interface receives the logoff message and performs the following steps:
 - a. Clears the following timers running for the co-network and its stations:
 - Network Management Message timers
 - Extended Network Management Message timers
 - Wait-for-Traffic timers
 - Performance timer
 - Key Exchange timers
 - Clear Old Key timersAll other timers are left running to ensure that current transactions in progress complete.
 - b. Marks all of the co-network's stations down.
 - c. Marks the co-network as down and logged off.
3. The co-network sends an echo test or logon message to reestablish communications.

Change Key Message Example

The diagram below illustrates the message flow between the co-network and BIC ANSI Interface when the co-network initiates a change key request message. In this case, the BIC ANSI Interface is responsible for generating the key being requested. The steps corresponding to the diagram are also described below.



1. The co-network formats and sends an 0800 message with a network management information code of 161 to the BIC ANSI Interface. This message includes the KEY-MGMT.X917-CSM-DATA field containing a Request Service Initiation Message (RSI).
2. The BIC ANSI Interface receives the 0800 message and validates it before responding. To validate the message, the BIC ANSI Interface performs as follows:
 - a. Verifies that the RSI is in the correct format. If the RSI is not in the correct format, the BIC ANSI Interface sets an error flag to F to indicate that a format error was found.
 - b. Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the BIC ANSI Interface compares the receiver ID from the RSI to the value from the ORIGINATING ID field on KEYF screen 3, and the originator ID from the RSI to the value from the RECEIVING ID field on KEYF screen 3.

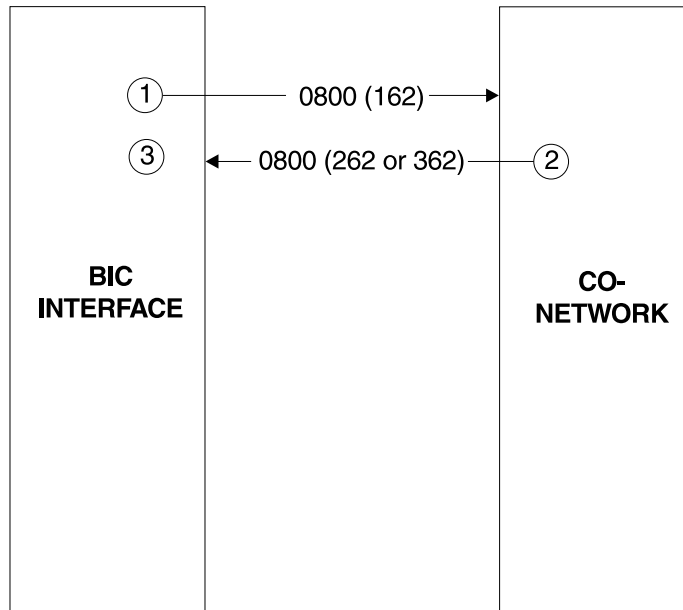
If the receiver or originator does not match the corresponding KEYF values for this co-network, the BIC ANSI Interface sets an error flag to H to indicate that the receiver or originator ID is incorrect.

- c. Verifies that there is not already a change key action in progress for the type of key identified in the RSI. The BIC ANSI Interface checks an internal flag to verify that a key exchange is not currently in progress. If a key change is in progress, the BIC ANSI Interface sets an error flag to C to indicate that the current change key request cannot be processed.
- d. Formats an 0800 response message and returns it to the co-network. To format the response message, the BIC ANSI Interface performs as follows:
 - If one or more of the above checks failed with an error, the BIC ANSI Interface formats an 0800 response message with the response code of 361. This message includes the KEY-MGMT.X917-CSM-DATA field containing an Error Service Message (ESM). The ERF field in the ESM includes any error flag values set during the checks. No further processing is performed.
 - If no errors were encountered in the above checks, the BIC ANSI Interface formats an 0800 response message with a network management information code of 261. This message includes the KEY-MGMT.X917-CSM-DATA containing a copy of the original RSI.
- e. Creates a new key and sends it to the co-network in a new key message.

For a description of this processing, refer to the “New Key Message Example” immediately following.

New Key Message Example

The diagram below illustrates the message flow between the BIC ANSI Interface and the co-network when the BIC ANSI Interface sends a new key message. The steps corresponding to the diagram are also described below.

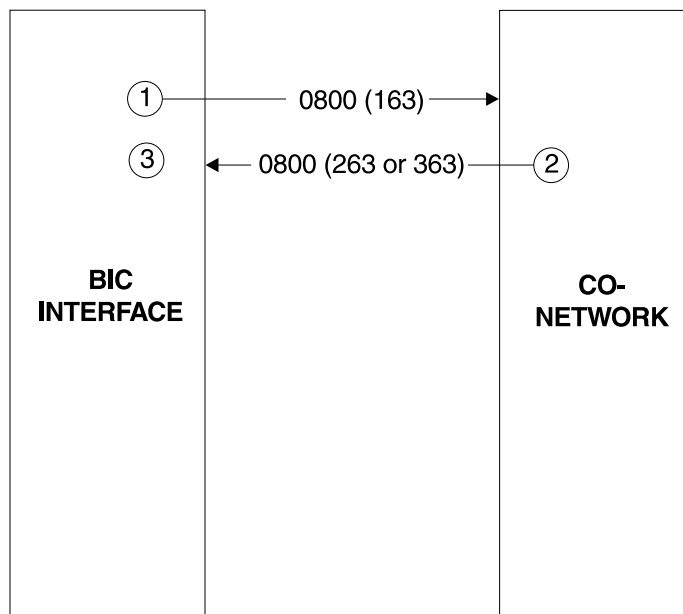


1. The BIC ANSI Interface creates a new key message and sends it to the co-network. To do this, the BIC ANSI Interface performs as follows:
 - a. Generates the new key and check digits.
 - b. Formats a Key Service Message (KSM) containing the new key and the check digits.
 - c. Sends an 0800 message to the co-network with the network management information code set to 162. This message includes the KEY-MGMT. X917-CSM-DATA containing the KSM.
 - d. Sets an internal flag indicating that a key exchange is in progress.
 - e. Starts a Network Management Message timer for the message. If this timer expires before an 0800 response is received from the co-network, the BIC ANSI Interface starts an Extended Network Management Message timer for the message. When the Extended Network Management Message timer expires, the BIC ANSI Interface resends the new key message.

2. The co-network receives the new key message, processes it, and returns an 0800 response message to the BIC ANSI Interface. This message includes the KEY-MGMT.X917-CSM-DATA field containing either a Response Service Message (RSM) or an ESM.
3. The BIC ANSI Interface receives the 0800 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out.
 - If the 0800 message timed out before the 0800 response was received, the BIC ANSI Interface has already set an Extended Network Management Message timer for the new key message (see step 1e). No further steps are performed.
 - If the 0800 message has not timed out, processing continues with the next step.
 - b. Clears the internal flag indicating that a key exchange is in progress.
 - c. Continues processing depending on the response code sent in the 0800 response message.
 - 1) If the transaction was denied (the network management information code was 362), the BIC ANSI Interface parses the KEY-MGMT.X917-CSM-DATA field to determine why the new key was denied. In this case, the KEY-MGMT.X917-CSM-DATA contains an ESM.
 - If the ESM indicates the message was denied because BASE24's key counter does not match the key counter maintained by the co-network, the BIC ANSI Interface resets its key counter in the KEYF to the value sent by the co-network. The BIC ANSI Interface then resends the new key message to the co-network. Processing returns to step 1c.
 - If the ESM indicates that any other error occurred, the BIC ANSI Interface sends a message indicating the error to the network logging facility and drops the 0800 response.
 - 2) If the transaction was approved (the network management information code was 262), the BIC ANSI Interface updates the KEYF with the new key, check digit, and counter information. The BIC ANSI Interface also reinitializes its internal key change threshold counters and timers.
 - d. The BIC ANSI Interface sets a clear old key timer to clear the old key.

Repeat Key Message Example

The diagram below illustrates the message flow between the BIC ANSI Interface and the co-network when the BIC ANSI Interface initiates a repeat key request message. In this case, the co-network is responsible for generating the key being requested. The BIC ANSI Interface will only initiate a repeat key message if it receives a KEY REPEAT command from the Network Control Supervisor (NCS). The processing described below starts from the point where the BIC ANSI Interface receives the NCS command. The steps corresponding to the diagram are also described below.



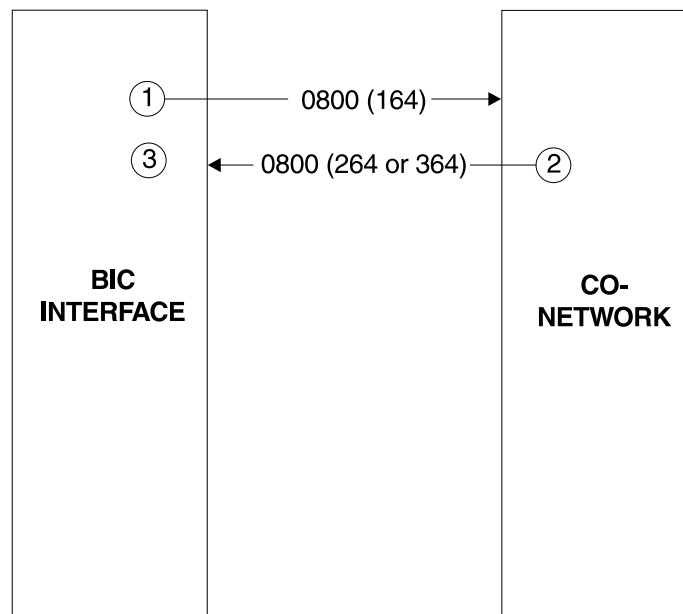
1. When the BIC ANSI Interface receives the REPEAT KEY command from the NCS, it performs as follows:
 - a. Verifies that the syntax is correct and that dynamic key management is supported. If the syntax is incorrect or dynamic key management is not supported, the BIC ANSI Interface logs a message and drops the command. If the syntax is correct and dynamic key management is supported, the BIC ANSI Interface continues with the next step (step 1b).

- b. Checks for an available station to which to send the message.
 - If no stations are available, the BIC ANSI Interface logs a message indicating that it is unable to send the repeat key message and drops the REPEAT KEY command. No further processing is performed.
 - If a station is available, processing continues with the next step (1c).
 - c. Sends an 0800 message to the co-network with the network management information code set to 163. This message includes a zero-filled KEY-MGMT.X917-CSM-DATA field.
 - d. Starts a Network Management Message timer for the message. If this timer expires before an 0800 response is received from the co-network, the BIC ANSI Interface logs a message indicating the repeat key message timed out and no further processing is performed.
2. The co-network receives the repeat key message, processes the message, and returns an 0800 response with KSM data to the BIC ANSI Interface.
 3. The BIC ANSI Interface receives the 0800 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out. If the 0800 message timed out before the 0800 response was received, the BIC ANSI Interface drops the message, and no further steps are performed. If the 0800 message has not timed out, processing continues with the next step.
 - b. Continues processing depending on the network management information code sent in the 0800 response message.
 - 1) If the transaction was denied (the network management information code was 363), the BIC ANSI Interface logs a message and drops the 0800 message.
 - 2) If the transaction was approved (the network management information code was 263), the BIC ANSI Interface performs as follows:
 - a) Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the BIC ANSI Interface compares the receiver ID from the RSI to the value from the ORIGINATING ID field on KEYF screen 3, and the originator ID from the KSM to the value from the RECEIVING ID field on KEYF screen 3. If the receiver or originator does not match the corresponding KEYF values for this co-network, the BIC ANSI Interface logs a message and drops the 0800 response.

- b) Verifies the key. To do this the BIC ANSI Interface translates the key sent by the co-network and verifies the check digits. The check digits generated as a result of translating the key must match the check digits sent by the co-network. If the BIC ANSI Interface cannot translate the key or the check digits do not match, the BIC ANSI Interface logs a message and drops the 0800 response.
- c) Updates the KEYF with the new key and check digit information.

Verify Key Request Example

The diagram below illustrates the message flow between the BIC ANSI Interface and the co-network when the BIC ANSI Interface initiates a verify key request message. In this case, the co-network is responsible for generating the key being verified. The BIC ANSI Interface process will only initiate a verify key message if it receives a KEY VERIFY command from the Network Control Supervisor (NCS). The processing described below starts from the point where the BIC ANSI Interface receives the NCS command. The steps corresponding to the diagram are also described below.



1. When the BIC ANSI Interface receives the VERIFY KEY command, it performs as follows:
 - a. Verifies that the syntax is correct and that dynamic key management is supported. If the syntax is incorrect or dynamic key management is not supported, the BIC ANSI Interface logs a message and drops the command. If the syntax is correct and dynamic key management is supported, the BIC ANSI Interface continues with the next step (step 1b).

- b. Checks for an available station to which to send the message.
 - If no stations are available, the BIC ANSI Interface logs a message indicating that it is unable to send the repeat key message and drops the REPEAT KEY COMMAND. No further processing is performed.
 - If a station is available, processing continues with the next step (step 1c).
 - c. Sends an 0800 message to the co-network with the network management information code set to 164. This message does not include the KEY-MGMT.X917-CSM-DATA field.
 - d. Starts a Network Management Message timer for the message. If this timer expires before an 0800 response is received from the co-network, the BIC ANSI Interface logs a message indicating that the verify key message timed out. No further processing is performed.
 2. The co-network receives the verify key request message, processes the message, and returns an 0800 response to the BIC ANSI Interface.
 3. The BIC ANSI Interface receives the 0800 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out. If the 0800 message timed out before the 0800 response was received, the BIC ANSI Interface drops the message, and no further steps are performed. If the 0800 message has not timed out, processing continues with the next step.
 - b. Continues processing depending on the response code sent in the 0800 response message.
 - If the transaction was denied (the network management information code was 364), the BIC ANSI Interface logs a message indicating the co-network denied the request and drops the 0800 message.
 - If the transaction was approved (the network management information code was 264), the BIC ANSI Interface logs a message indicating that the command was successful and drops the 0800 message.

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Section 4

Transaction Handling

This section describes how the BIC ANSI Interface process handles various types of transactions. It describes the basic flow of transaction processing and provides transaction flow diagrams illustrating how messages to and from the co-network are handled. Transaction flow diagrams are also provided for network management and settlement messages.

Basic Transaction Flow

When the BIC ANSI Interface receives a message, the basic flow is to process the message and then wait for another message. The processing performed on a message depends on the kind of message received. The transaction flows in this section diagram the flow of messages between BASE24 and the co-network. The following transaction flows are described in this section:

- Requests
- Reversals
- Timeouts
- Force Posts
- Store-and-Forward
- Logon
- Logoff
- Echo Test
- Settlement

In most cases, these flows are described twice—once when the message flow is initiated by BASE24 and once when the message flow is initiated by the co-network.

Logging to the ILF and SAF

The following messages handled by the BIC ANSI Interface are logged to the Interchange Log File (ILF):

- Requests
- Reversals
- Force posts
- Settlement

The following messages are logged to the Store-and-Forward File (SAF) if the communications link between BASE24 and the co-network is down:

- Reversals
- Force posts
- Settlement
- Cutover

Trace Numbers

When the BIC ANSI Interface receives a financial transaction request, it assigns a trace number to the transaction. The trace number is sent with the transaction to the appropriate authorizer, whether it is a BASE24 Authorization process or an authorization process in the co-network. When the transaction response is returned, the trace number is included as part of the response message. The trace number is used by the BIC ANSI Interface to look up the original request in memory that has been waiting on a reply. Trace numbers are also included in the information logged to the ILF for each financial transaction.

Likewise, the co-network must assign a trace number to each financial transaction request it receives. BASE24 differentiates these trace numbers by placing them in separate fields in both the internal and external message. One field contains the *acquirer* trace number—the trace number assigned by the entity acting on behalf of the transaction acquirer. Another field contains the *issuer* trace number—the trace number assigned by the entity acting on behalf of the card-issuing financial institution. Depending on where the transaction originated, the BIC ANSI Interface assigns either the acquirer trace number or the issuer trace number and the co-network assigns the other trace number.

Both the acquirer trace number and the issuer trace number must be saved in order to look up the original transaction in the event of a reversal. In order for the transaction acquirer to supply the issuer with the issuer's trace number, the acquirer uses the acquirer trace number to retrieve the record from its log file. Once the record has been retrieved from the acquirer's log file, the issuer's trace number, which was stored as part of the external message, is placed in the reversal message. This enables the issuer to look up the message using its own trace number.

Each ILF has its own series of trace numbers, starting at zero. Since the trace number is used as an alternate key to the file, trace numbers cannot be duplicated during a single processing day. If the BIC ANSI Interface process is reinitialized in the middle of the processing day, the ILF is read to determine the last trace number used. This number is then increased by 100 (to allow for outstanding transactions with trace numbers already assigned) and stored in the Process Control Table (PCT). The trace number is incremented every time a new trace number is assigned.

Message Flows to the Co-Network

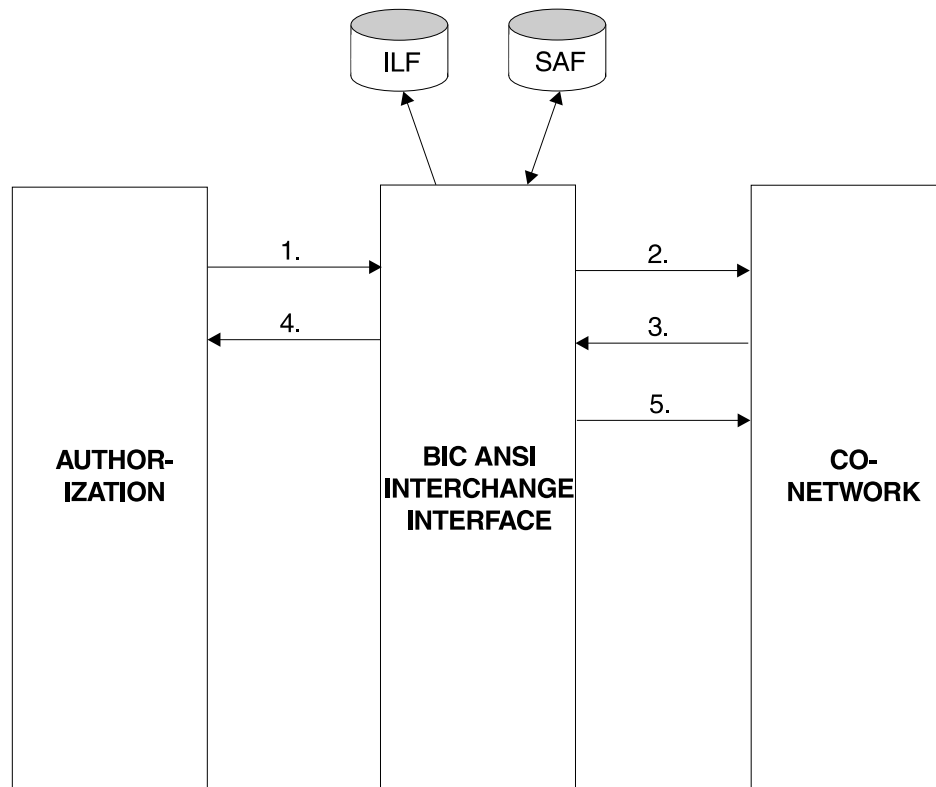
When a transaction is initiated on BASE24 that must be sent to the co-network for authorization, the following possible message flows can occur between BASE24 and the co-network:

- Requests
- Reversals
- Force Posts
- Timeouts
- Store-and-Forward

Each of these message flows is described on the following pages.

Request Message to the Co-Network

When a cardholder from the co-network initiates a transaction on the local BASE24 network, the transaction must be sent to the co-network for authorization. The diagram below illustrates message routing when a transaction is sent to the co-network for authorization.



The steps below describe the processing involved in the illustration:

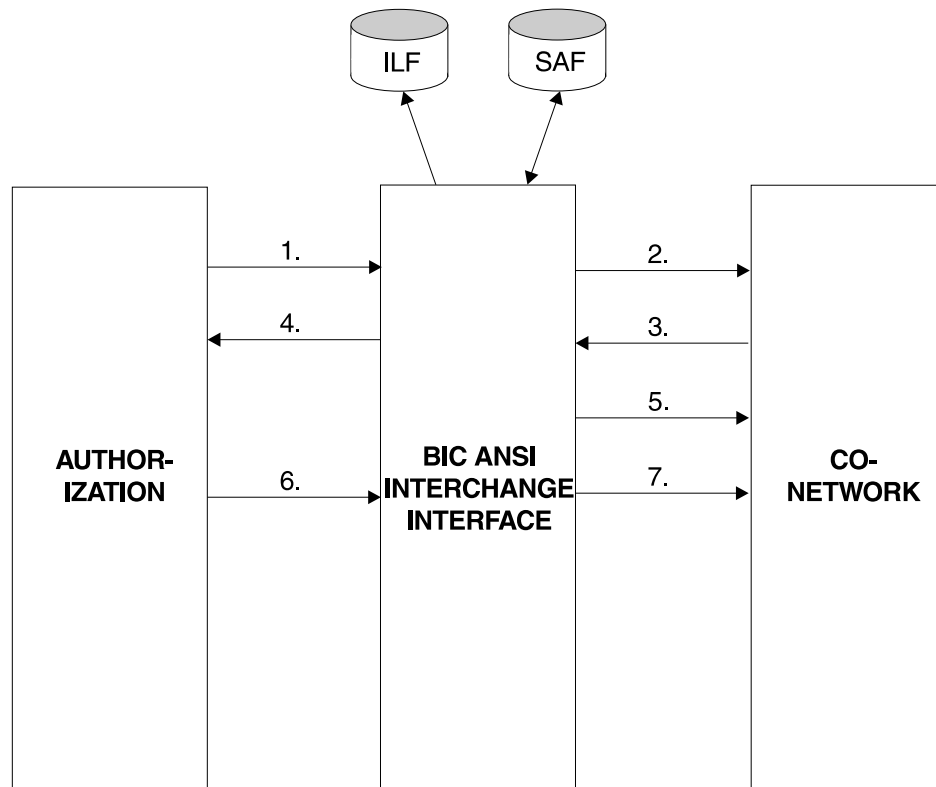
1. A BASE24 Authorization process sends the request to the BIC ANSI Interface in the 0200 internal message format.
2. The BIC ANSI Interface reformats the message into the 0200 or 0700 external message format and sends the external message to the co-network for authorization. An outbound request timer is set to wait for a response from the co-network.
3. The co-network routes the transaction to the appropriate authorizer. After the transaction has been authorized, the response is formatted into the 0210 or 0710 external message format and sent to the BIC ANSI Interface.

4. The BIC ANSI Interface deletes the outbound request timer and formats a 0210 internal response message. The response is returned to BASE24 to be completed at the ATM or POS device. A copy of the transaction is logged to the Interchange Log File (ILF).
5. If the co-network elected to have completions sent and the request was approved, the BIC ANSI Interface formats a 0221 external completion message. If the BIC ANSI Interface supports completion acknowledgments, the completion is added to the Store-and-Forward File (SAF) to be processed during regular SAF processing.

If completion acknowledgments are not being used, the transaction is sent to the co-network.

Reversal to the Co-Network

When an ATM or POS device connected to the local BASE24 network fails to complete a transaction as authorized, a reversal message must be sent to the co-network. The diagram below illustrates message routing when a reversal is sent to the co-network.



The steps below describe the processing involved in the illustration:

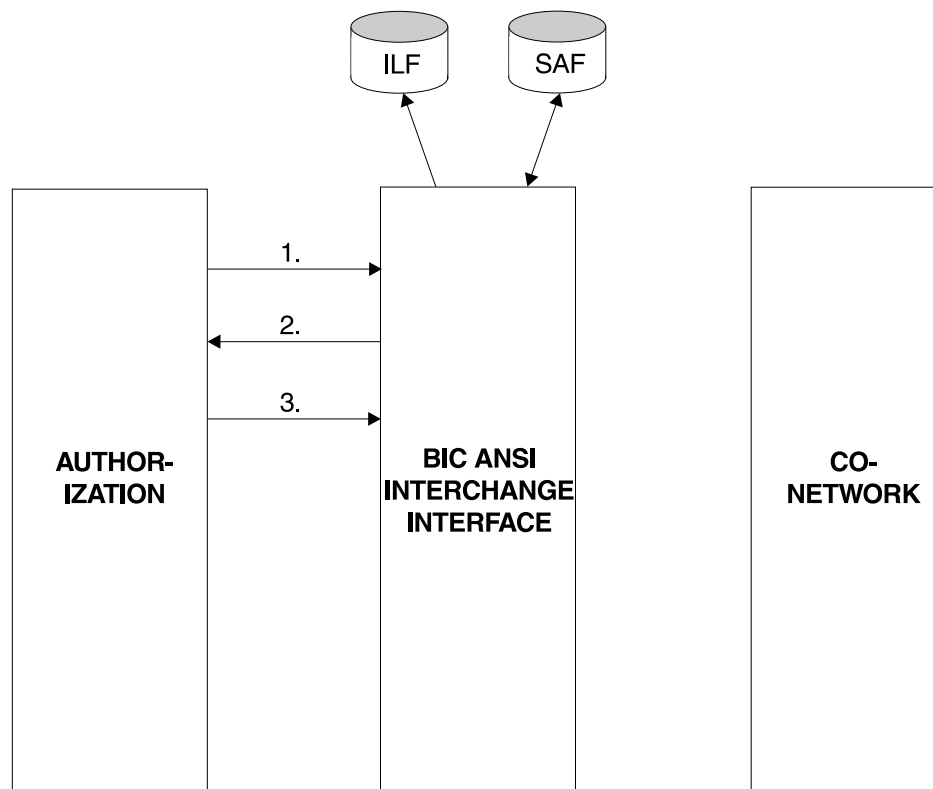
1. A BASE24 Authorization process sends the request to the BIC ANSI Interface in the 0200 internal message format.
2. The BIC ANSI Interface reformats the message into the 0200 or 0700 external message format and sends the external message to the co-network for authorization. An outbound request timer is started to wait for a response from the co-network.
3. The co-network routes the transaction to the appropriate authorizer. The response is formatted into the 0210 or 0710 external message format and sent to the BIC ANSI Interface.

4. The BIC ANSI Interface deletes the outbound request timer and formats a 0210 internal response message. The response is returned to BASE24 to be completed at the ATM or POS device. A copy of the transaction is logged to the Interchange Log File (ILF).
5. If the co-network elected to have completions sent and the request was approved, the BIC ANSI Interface formats a 0221 external completion message. If the BIC ANSI Interface supports completion acknowledgments, the completion is added to the Store-and-Forward File (SAF) to be processed during regular SAF processing.

If completion acknowledgments are not being used, the transaction is sent to the co-network.
6. The BASE24 Authorization process returns a 0420 message to the BIC ANSI Interface, indicating that the transaction failed to complete as authorized. The BIC ANSI Interface locates the ILF record for the original transaction using the trace number it assigned to the transaction and updates the ILF record with the reversal information.
7. The BIC ANSI Interface formats a 0400 external message. If the co-network elected not to have reversals after completions suppressed, the external message is logged to the SAF to be processed by the co-network during SAF processing. If the co-network elected to suppress reversals after completions, the external message is logged to the ILF and is not added to the SAF.

Force Post Messages to the Co-Network

When the local BASE24 network performs stand-in authorization, it sends the co-network a force post message for each transaction it authorizes. The diagram below illustrates message routing when a force post message is sent to the co-network.

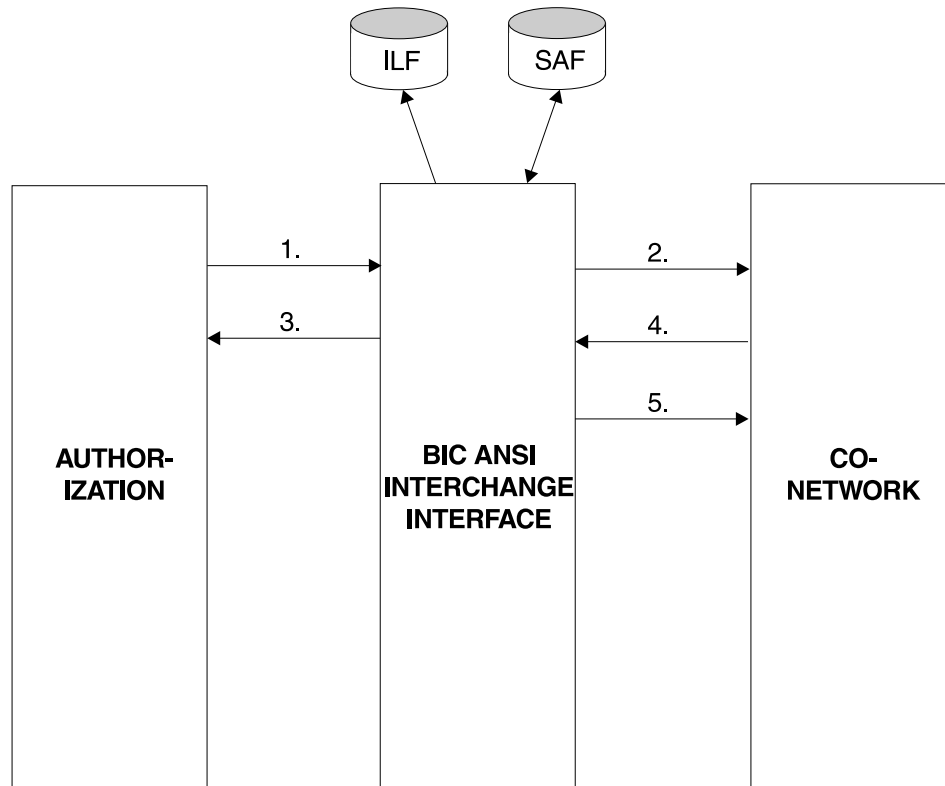


The steps below describe the processing involved in the illustration:

1. A BASE24 Authorization process sends the request to the BIC ANSI Interface in the 0200 internal message format.
2. The BIC ANSI Interface determines that the co-network is not available and sends the message back to BASE24 to be authorized.
3. The BASE24 Authorization process sends the BIC ANSI Interface a force post message in the 0220 internal message format. If the force post was approved, the BIC ANSI Interface reformats the message into the 0220 external message format and adds the external message to the Store-and-Forward (SAF). A copy of the transaction is added to the Interchange Log File (ILF).

Timeouts of Requests to Co-Network

Each time the BIC ANSI Interface sends a 0200 financial transaction request to the co-network for authorization, the BIC ANSI Interface sets an outbound request timer. If the outbound request timer expires before a response is received from the co-network, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out.



The steps below describe the processing involved in the illustration:

1. A BASE24 Authorization process sends the request to the BIC ANSI Interface in the 0200 internal message format.
2. The BIC ANSI Interface reformats the message into the 0200 external message format and sends the external message to the co-network for authorization. An outbound request timer is set to wait for the response from the co-network.
3. When the timer expires before a response is received from the co-network, the BIC ANSI Interface checks for an alternate destination. If an alternate destination does not exist, the BIC ANSI Interface determines whether stand-in authorization is allowed. If stand-in authorization is allowed, the

transaction is returned to BASE24 as a 0200 message. If stand-in authorization is not allowed, the BIC ANSI Interface formats a 0210 internal response message denying the transaction. The response is returned to BASE24. A copy of the transaction is logged to the Interchange Log File (ILF).

4. The co-network sends a 0210 late response to the BIC ANSI Interface. If the transaction was denied by the co-network, the BIC ANSI Interface simply drops the late response.
5. If the transaction was approved by the co-network, the BIC ANSI Interface updates the ILF record for the transaction. The BIC ANSI Interface formats a 0400 external reversal message and adds it to the Store-and-Forward (SAF) to be processed during store-and-forward processing.

Store-and-Forward Messages to the Co-Network

Messages intended for the co-network are stored in the Store-and-Forward File (SAF) during periods when the communications link between BASE24 and the co-network is down. In addition, completion and adjustment messages are regularly logged to the SAF. Periodically, the BIC ANSI Interface checks to determine whether a SAF record should be sent. If the link to the co-network is up, the SAF contains records, and there is not a SAF record in the process of being sent to the co-network, the BIC ANSI Interface retrieves the first record from the SAF and sends it to the co-network. Only one message from the SAF is sent at a time. The general processing flow for SAF messages is the same as for normal messages. To the co-network, the messages appear to be happening as they are sent—they cannot be distinguished from normal messages. The BIC ANSI Interface uses different timers that identify the messages as SAF records.

The following message types can be logged to the SAF:

- 0220 Force Post messages
- 0221 Completion messages
- 0400 Reversal messages
- 0500 Settlement Totals messages
- 0800 Cutover messages
- 5400 Adjustment messages

The co-network must send any required acknowledgments to these messages to the BIC ANSI Interface.

Message Flows from the Co-Network

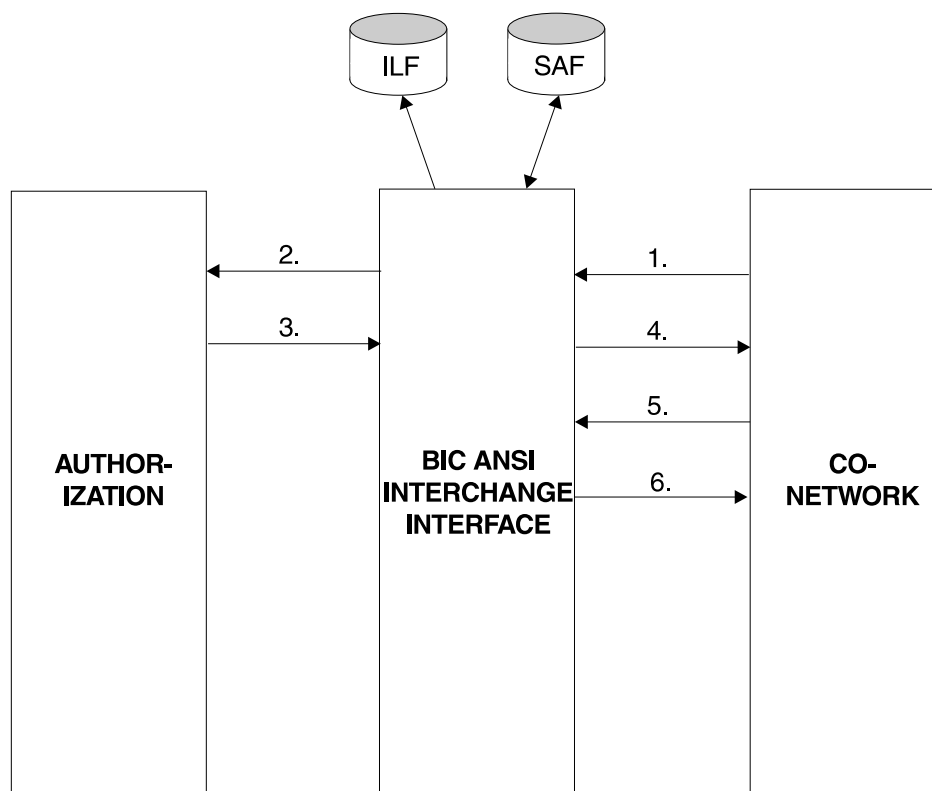
When a transaction is initiated on the co-network that must be authorized by BASE24, the following possible message flows can occur between BASE24 and the co-network:

- Requests
- Reversals
- Timeouts

Each of these message flows is described on the following pages.

Request Message from the Co-Network

When a cardholder from the local BASE24 network initiates a transaction on the co-network, the transaction must be sent to BASE24 for authorization. The diagram below illustrates message routing when a transaction is sent to BASE24 for authorization.

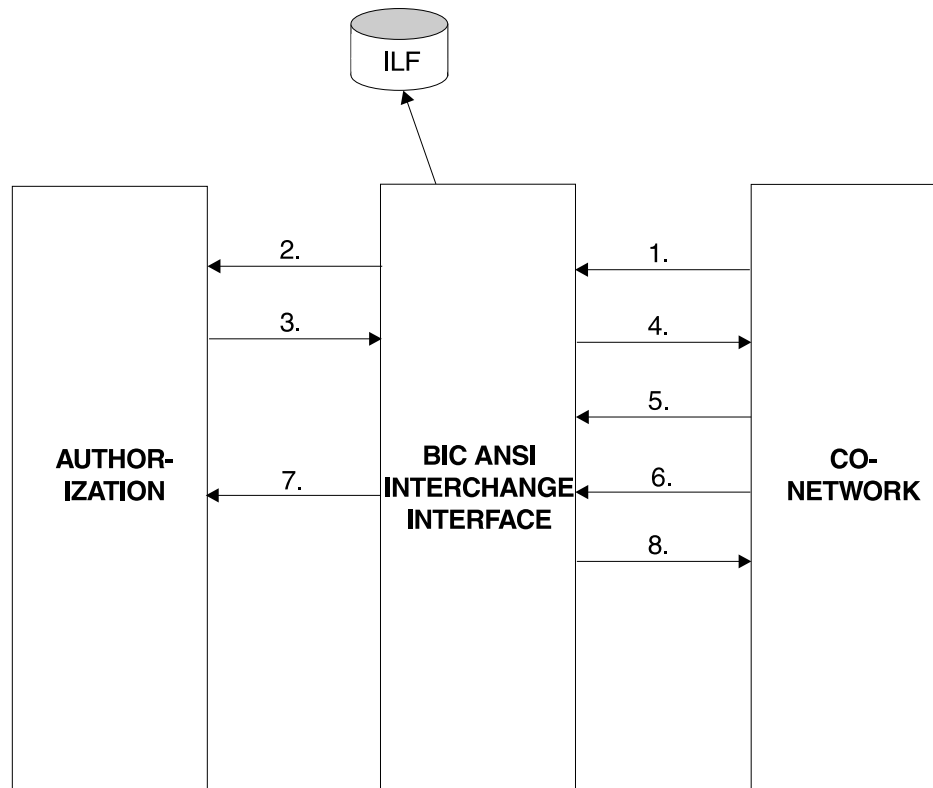


The steps below describe the processing involved in the illustration:

1. The co-network sends the request to the BIC ANSI Interface in the 0200 or 0700 external message format.
2. The BIC ANSI Interface reformats the message into the 0200 internal message format and sends the internal message to BASE24 for authorization. An inbound request timer is started to wait for the response.
3. The BASE24 Authorization process sends the response to the BIC ANSI Interface in the 0210 internal message format.
4. The BIC ANSI Interface deletes the inbound request timer and formats a 0210 or 0710 external response message. The response is returned to the co-network to be completed. A copy of the transaction is logged to the Interchange Log File (ILF).
5. If the transaction was approved and completions are required, the co-network sends a 0221 external completion message to the BIC ANSI Interface.
6. If completion acknowledgments are required, the BIC ANSI Interface sends a 0230 completion acknowledgment to the co-network.

Reversal from the Co-Network

When an ATM or POS device connected to the co-network fails to complete a transaction as authorized, a reversal message must be sent to BASE24. The diagram below illustrates message routing when a reversal is sent to BASE24.



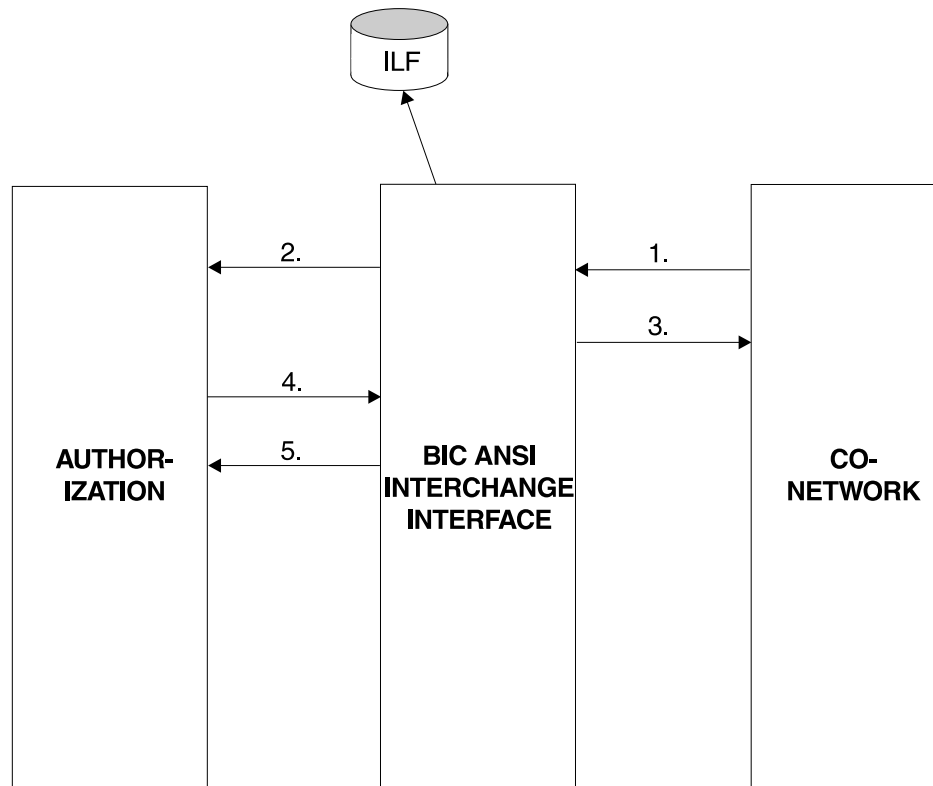
The steps below describe the processing involved in the illustration:

1. The co-network sends the request to the BIC ANSI Interface in the 0200 external message format.
2. The BIC ANSI Interface reformats the message into the 0200 internal message format and sends the internal message to BASE24 for authorization. An inbound request timer is started to wait for a response from BASE24.
3. The BASE24 Authorization process sends the transaction response to the BIC ANSI Interface in the 0210 internal message format.
4. The BIC ANSI Interface deletes the inbound request timer and formats a 0210 external response message. The response is returned to the co-network to be completed at the ATM or POS device. A copy of the transaction is logged to the Interchange Log File (ILF).

5. If completions are required and the request was approved, a 0221 external completion message is received from the co-network.
6. The co-network returns a 0400 or 402 message to the BIC ANSI Interface, indicating that the transaction failed to complete as authorized.
7. The BIC ANSI Interface updates the ILF record for the original transaction and formats a 0420 internal message. The internal message is sent to BASE24.
8. If the reversal acknowledgment suppression option is not being used, a 0410 or 0412 external reversal acknowledgment is sent to the co-network.

Timeouts of Request Messages from the Co-Network

Each time the BIC ANSI Interface sends a financial transaction to a BASE24 Authorization process for authorization, the BIC ANSI Interface sets an inbound request timer. If the inbound request timer expires before a response is received from BASE24, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out.



The steps below describe the processing involved in the illustration:

1. The co-network sends the request to the BIC ANSI Interface in the 0200 or 0700 external message format.
2. The BIC ANSI Interface reformats the message into the 0200 internal message format and sends the internal message to BASE24 for authorization. An inbound request timer is set to wait for a response from the Authorization process.
3. When the inbound request timer expires, the BIC ANSI Interface formats a 0210 or 0710 external response message denying the transaction and returns it to the co-network. A copy of the transaction is logged to the Interchange Log File (ILF).

4. The BASE24 Authorization process sends a late 0210 response to the BIC ANSI Interface. If the transaction was denied by the BASE24 Authorization process, the BIC ANSI Interface simply drops the late response.
5. If the transaction was approved, the BIC ANSI Interface formats a 0420 internal reversal message and sends the reversal to BASE24.

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Section 5

Cutover and Reporting

This section describes BIC ANSI Interface cutover processing and report generation. It includes detailed descriptions of the current cutover message flows. This section also provides samples of the reports generated for the BIC ANSI Interface.

Introduction

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports are provided to aid in reconciling the BASE24 network with the co-network. Cutover processing occurs seven days a week.

Interchange cutover is initiated by the BASE24-atm or BASE24-pos Settlement Initiator process. As part of product cutover, the Settlement Initiator process sends the BIC ANSI Interface a cutover message. If only one BASE24 product is present, interchange cutover occurs at the same time as BASE24 cutover.

If both BASE24-atm and BASE24-pos are present, the BIC ANSI Interface process determines when cutover should occur by comparing the cutover times set in the LCONF for BASE24-atm and BASE24-pos. For example, if BASE24-atm cuts over at 3:00 p.m. and BASE24-pos cuts over at 9:00 p.m., the BASE24-atm Settlement Initiator process sends the BIC ANSI Interface process a cutover message at 3:00 p.m. When the BIC ANSI Interface process receives the cutover message, it updates the BASE24-atm processing date and compares the two cutover times. Since BASE24-atm does not have the last cutover time for the day, the interchange is not cutover. At 9:00 p.m., the BASE24-pos Settlement Initiator process sends the BIC ANSI Interface a cutover message. When the BIC ANSI Interface process receives the cutover message, it updates the BASE24-pos processing date and compares the two cutover times. Since BASE24-pos has the last cutover time, the interchange is also cutover.

BIC ANSI Totals

The BIC ANSI Interface keeps a number of running totals that reflect the transactions processed between BASE24 and the co-network. The totals kept by the BIC ANSI Interface include the following:

- Gross debit amount
- Gross debit count
- Gross credit amount
- Gross credit count
- Reversal debit amount
- Reversal debit count
- Reversal credit amount
- Reversal credit count

These totals are kept for the current processing date, the previous processing date, and the next processing date. As each transaction is processed by the BIC ANSI Interface, the appropriate totals are updated based on the type of transaction and the posting date for the transaction. The totals kept by the BIC ANSI Interface are sent to the co-network in the 0500 Settlement Totals message and are used by the Interchange report programs.

Cutover Message Processing

The following paragraphs describe the processing performed by the BIC ANSI Interface process when it receives a cutover message from BASE24 or the co-network.

BASE24 Settlement Initiator Process Sends Cutover Message

The following steps are taken by the BIC ANSI Interface process when a cutover message is received from a BASE24 Settlement Initiator process. When the cutover message is received, the BIC ANSI Interface process performs as follows:

1. Updates the current business date for the product whose Settlement Initiator process sent the cutover message.
2. Compares the cutover times found in the LCONF for BASE24-pos and BASE24-atm to determine if the BIC ANSI Interface should be cutover at this time. BIC ANSI Interface cutover occurs at the later of the two cutover times.

If the BIC ANSI Interface should not be cut over at this time, no further cutover steps are taken.

3. Checks to see if cutover totals have already been sent for this date by comparing the date sent in the cutover message with the previous cutover date the BIC ANSI Interface keeps in memory.

If the totals have already been sent, the BIC ANSI Interface has already been cut over and no further cutover steps are taken.

4. Updates the settlement totals that are kept in memory. To do this, the BIC ANSI Interface performs as follows:
 - a. Sets the previous day totals to equal the current day totals
 - b. Sets the current day totals to equal the next day totals
 - c. Initializes the next day totals to zero.
5. Updates the settlement totals dates that are kept in memory. To do this, the BIC ANSI Interface performs as follows:
 - a. Sets the previous settlement totals date to the current processing date.
 - b. Initializes the next settlement totals date to blanks.

6. Checks the Interchange Configuration File (ICF) information stored in the Process Control Table (PCT) to determine whether this BIC ANSI Interface process is a slave to the co-network.

The slave network is not responsible for sending a cutover message to the co-network. If this process is a slave, no further cutover steps are taken.

7. Formats a cutover message and sends it to the co-network.
8. Determines whether this BIC ANSI Interface process is the master of the co-network.

If this BIC ANSI Interface process is not the master of the co-network, equal partner settlement has been performed and no further steps are required.

If this is the master network, the BIC ANSI Interface process opens a new Interchange Log File (ILF) and begins logging transaction records to the new ILF.

Co-Network Sends Cutover Message

The following steps are taken by the BIC ANSI Interface process when a cutover message is received from the co-network. When the cutover message is received, the BIC ANSI Interface process performs as follows:

1. If BASE24 is the master network, no further steps are taken.
2. If BASE24 is the slave network, sets a timer to wait for the settlement totals message and continues with the next step.
3. Determines if a cutover message has already been processed for the co-network's processing date by comparing the date from the external message with the co-network's posting date kept in the PCT. If so, a message is logged and no further cutover steps are taken.
4. Updates the current business date for the co-network.
5. Opens and begins logging transaction records to a new ILF.
6. Sends a cutover acknowledgment message to the co-network.

Cutover Message Flows

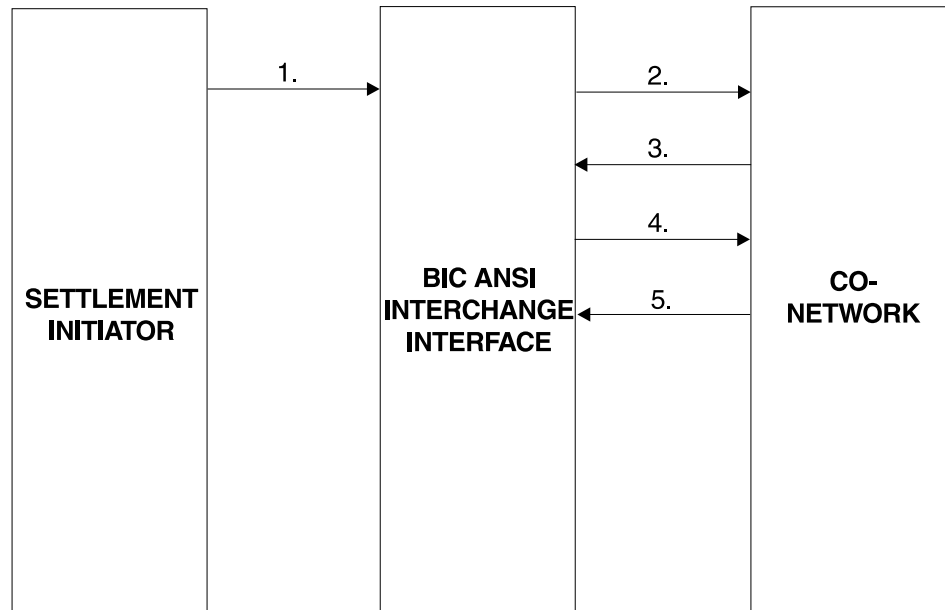
Cutover processing occurs seven days a week. The BIC ANSI Interface keeps an ILF, which reflects the transaction activity for the co-network's business day. For example, when the co-network cuts over to a new business day, the BIC ANSI Interface closes its ILF and opens a new ILF reflecting the current business date of the co-network.

When both BASE24 and the co-network are equal partners, each network informs the other network when cutover occurs. In addition, each network must send the other network settlement totals.

When both networks are not equal partners, BASE24 must be identified as either the master network or the slave network in the ICF. The co-network is either master or slave, whichever BASE24 is not. Slave network cutover must occur before the master network cutover or incoming transactions from the slave network is denied. The master network must send cutover notification and settlement totals to the co-network when it cuts over. In this case, the co-network is not responsible for sending its own cutover notification and settlement totals.

Equal Partner Settlement/BASE24 Cutover

When BASE24 and the co-network are equal partners regarding settlement and cutover, the same processing occurs for both. The diagram below illustrates the settlement message flow when BASE24 cuts over.



The steps below describe the processing involved in the illustration.

1. The BIC ANSI Interface receives notification of cutover from a Settlement Initiator process in the BASE24 network.

Both BASE24-atm and BASE24-pos Settlement Initiator processes send cutover notification to the BIC Interchange. When the BIC ANSI Interface receives a cutover message from a BASE24 Settlement Initiator process, it updates the current processing date for the product whose Settlement Initiator process sent the message. Then, the BIC ANSI Interface checks to determine whether another BASE24 product has a later cutover. If another product has a later cutover, no further steps are taken at this time.

If the cutover message came from the Settlement Initiator process with the latest product cutover, the BIC ANSI Interface initiates Interchange Cutover.

2. The BIC ANSI Interface sends an 0800 type message with the information code set to 201 to the co-network. This 0800 message contains the new business date.
3. The co-network sends an 0800 cutover acknowledgment message to the BIC ANSI Interface using an information code of 601.

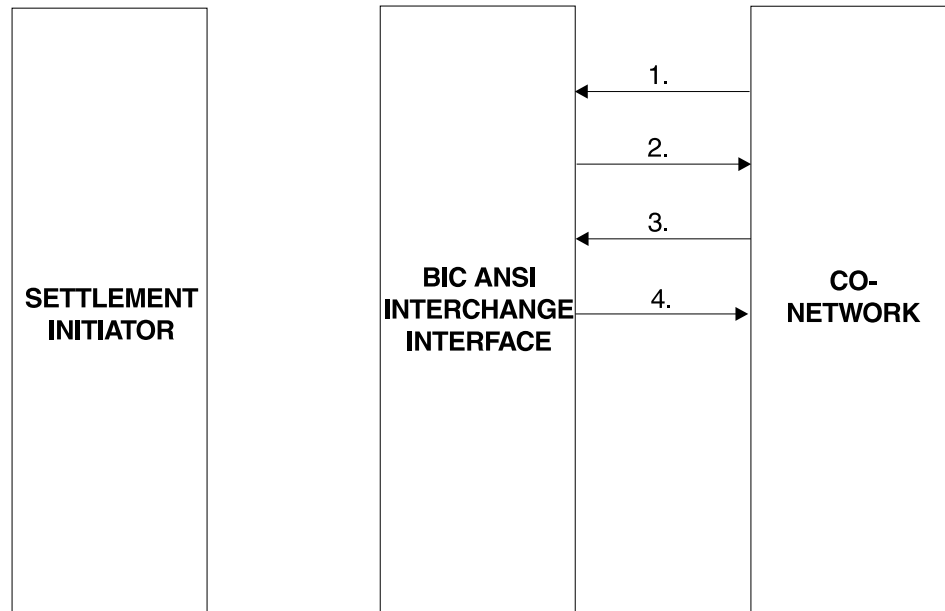
4. The BIC ANSI Interface logs a log message stating that the cutover acknowledgment message has been received. After waiting five minutes for all of the old business day's transactions to finish, the BIC ANSI Interface sends a type 0500 settlement totals message to the co-network and another log message is sent to the current log device stating that the settlement totals were sent.

The totals message contains totals for all shared activity for the BASE24 date that is being cutover.

5. The co-network replies with a type 0510 settlement totals acknowledgment message.

Equal Partner Settlement/Co-Network Cutover

The diagram below illustrates the settlement message flow when the co-network cuts over.



The steps below describe the processing involved in the illustration.

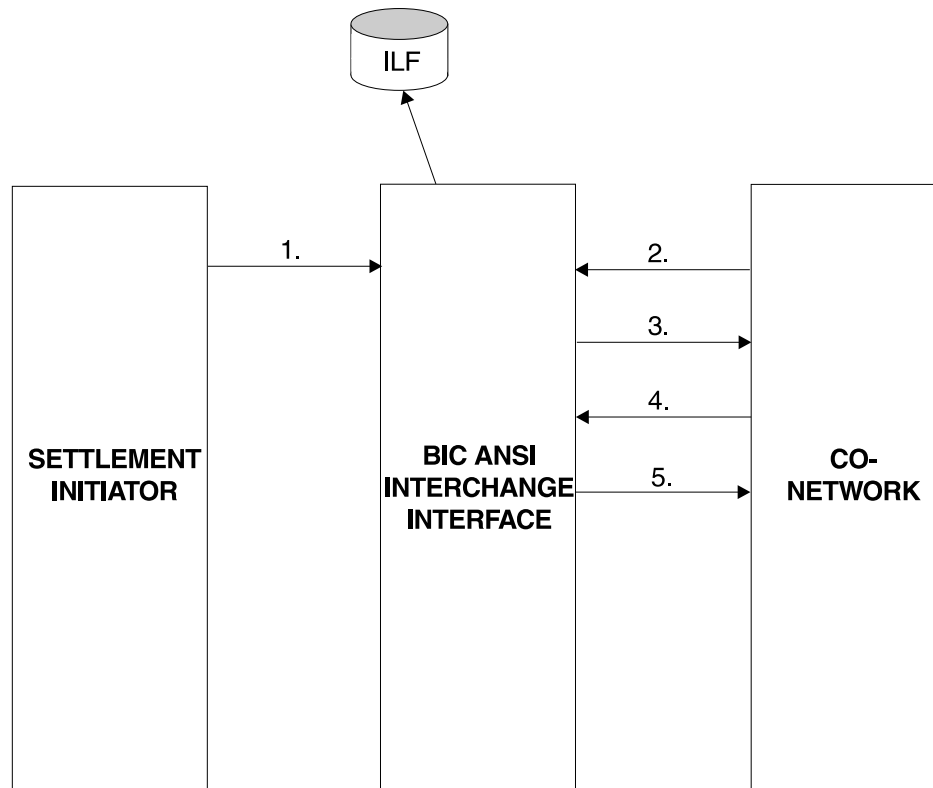
1. The co-network sends an 0800 type message with information code set to 201 to the BIC ANSI Interface. This 0800 message contains the new business date.
2. The BIC ANSI Interface opens a new Interchange Log File (ILF) and sends an 0800 cutover acknowledgment message to the co-network using an information code of 601.
3. After waiting for all of the old business day's transactions to finish, the co-network sends a type 0500 settlement totals message to the BIC ANSI Interface. The BIC ANSI Interface adds the totals message to the ILF for the business day just closed.

The totals message contains totals for all shared activity for the co-network date that is being cutover.

4. The BIC ANSI Interface replies with a type 0510 settlement totals acknowledgment message.

Master/Slave Settlement when BASE24 is Slave

When the two networks are not equal partners regarding settlement and cutover, the slave network must cutover before the master network. The diagram below shows the message flow when BASE24 is the slave network.



The steps below describe the processing involved in the illustration.

1. The BIC ANSI Interface receives notification of cutover from a Settlement Initiator process in the BASE24 network.

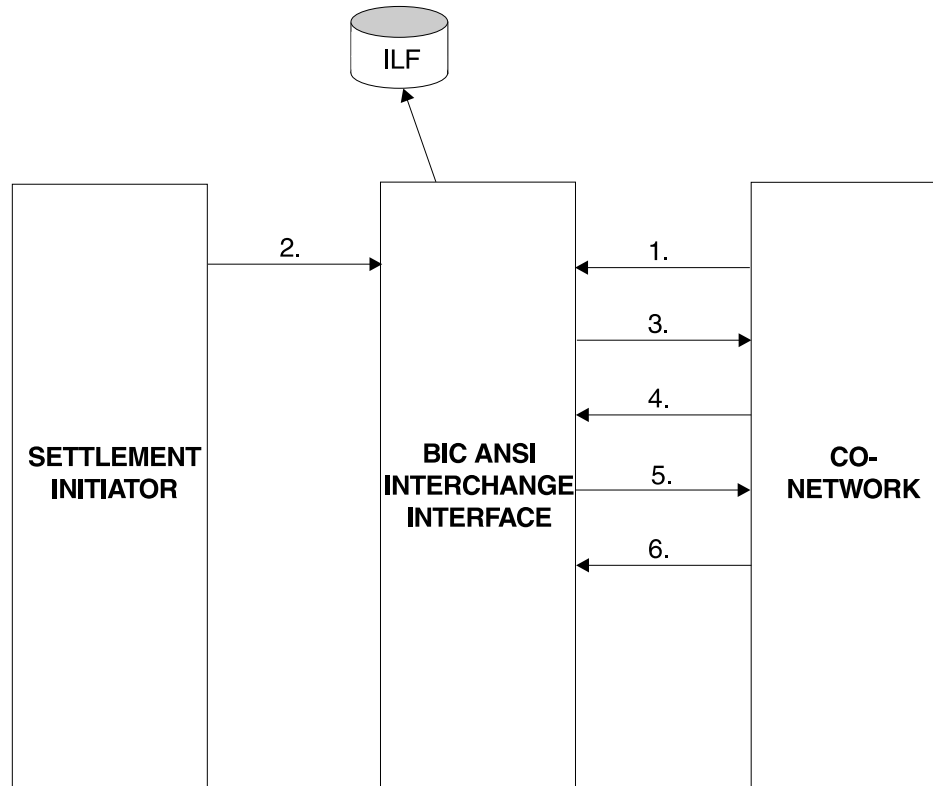
Both BASE24-atm and BASE24-pos Settlement Initiator processes send cutover notification to the BIC Interchange. When the BIC ANSI Interface receives a cutover message from a BASE24 Settlement Initiator process, it updates the current processing date for the product whose Settlement Initiator process sent the message. Then the BIC ANSI Interface checks to determine whether another BASE24 product has a later cutover. If another product has a later cutover, no further steps are taken at this time.

If the cutover message came from the Settlement Initiator process with the latest product cutover, the BIC ANSI Interface updates its totals and processing date.

2. At some later time, the co-network cuts over. The co-network sends a type 0800 message to the BIC ANSI Interface containing its new business date.
3. The BIC ANSI Interface opens a new Interchange Log File (ILF), if one has not already been opened, and replies to the 0800 cutover message with an 0800 acknowledgment.
4. After waiting for all of the old business date's transactions to complete, the co-network sends a type 0500 settlement totals message to the BIC ANSI Interface.
5. The BIC ANSI Interface logs the totals to the ILF for the business day just closed and acknowledges the 0500 message with a type 0510 message.

Master/Slave Settlement when BASE24 is Master

The diagram below shows the message flow when BASE24 is the master network.



The steps below describe the processing involved in the illustration.

1. The co-network cuts over. At some point after cutover, a 0200 financial request message is sent to the BIC ANSI Interface containing the new business date. The BIC ANSI Interface opens a new Interchange Log File (ILF) reflecting the current business date of the co-network and updates the co-network processing date stored in memory.
2. The BIC ANSI Interface receives notification of cutover from a Settlement Initiator process in the BASE24 network.

Both BASE24-atm and BASE24-pos Settlement Initiator processes send cutover notification to the BIC Interchange. When the BIC ANSI Interface receives a cutover message from a BASE24 Settlement Initiator process, it updates the current processing date for the product whose Settlement Initiator process sent the message. Then the BIC ANSI Interface checks to determine whether another BASE24 product has a later cutover. If another product has a later cutover, no further steps are taken at this time.

3. If the cutover message came from the Settlement Initiator process with the latest product cutover, the BIC ANSI Interface sends a type 0800 message to the co-network containing its new business date.
4. The co-network replies to the 0800 cutover message with an 0800 acknowledgment.
5. The BIC ANSI Interface sends a log message to the current log device stating cutover acknowledgment has been received. After waiting five minutes for all of the old business date's transactions to complete, the BIC ANSI Interface sends a type 0500 settlement totals message to the co-network. Another log message is logged stating the settlement totals were sent.
6. The co-network acknowledges the 0500 message with a type 0510 message.

Starting Reports

The BIC ANSI Interface is responsible for starting the interchange reports. It does so by creating an obey file to start the report programs and then running the obey file at the appropriate time.

Using the time specified in the Logical Network Configuration File (LCONF) param SW-RPT-STARTUP, the BIC ANSI Interface sets a timer to expire when the reports are to be run. When the timer expires, the BIC ANSI Interface creates and runs an obey file containing the commands necessary to run the report programs via Tandem Advanced Command Language (TACL).

Under normal circumstances, the report programs start and stop without operator intervention. However, if the BIC ANSI Interface does not start the programs, they can be started manually via TACL by running the report obey file created by the BIC ANSI Interface.

If the report programsabend or fail, they can be restarted in the same way.

Report Overview

The BIC ANSI Interface produces two reports of interchange transaction activity. These reports are the SETL01-BIC Clearings Statement Report for BIC and the STAT09-BIC BIC Transaction Detail Report. These reports can be used in conjunction with BASE24-atm and BASE24-pos reports to reconcile the differences between the BASE24 processing day and the co-network processing day.

This section provides an overview of the information provided in the interchange reports. For more detailed information and sample report formats, see the ***BASE24-atm Settlement and Reporting Manual*** or the ***BASE24-pos Settlement and Reporting Manual***. These manuals also contain information on using the interchange reports to effect settlement.

SETL01-BIC Clearings Statement Report for BIC

The Clearings Statement Report for BIC reflects the debit, credit, and net position of the co-network to BASE24. The amounts shown on the report are from the co-network's point of view, and so the debit and credit positions are reversed.

This report contains the totals of all transactions logged to the ILF. The first part of the report gives transaction totals for BASE24's processing day.

Following the BASE24 totals, there is a section for transactions that occurred during BASE24's processing day but that did not occur during the co-network processing day. These transactions are subtracted from the transaction totals for BASE24's processing day. This section provides two pieces of information: the net amount of transactions and the dollar difference between transactions occurring during BASE24's processing day but not during the co-network's processing day.

Finally, there is a section showing transactions that occurred during the co-network's processing day but that did not occur during BASE24's processing day. These transactions are added to the transaction totals for BASE24's processing day. This section provides two pieces of information: the net amount of transactions and the dollar difference between transactions occurring during the co-network's processing day but not during BASE24's processing day.

STAT09-BIC BIC Transaction Detail Report

The BIC Transaction Detail Report provides a complete listing of all interchange activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the BIC ANSI Interface during its processing day. The information provided for each transaction includes the cardholder number, transaction type, transaction time, posting date, and all transaction amounts.

Appendix A

BIC ANSI Interface External Message

The BIC external message, or SEM, is a variable-length, variable-content message that is configured differently based on the type of message being sent. This appendix describes the BIC external message.

External Message Structure

Each message that can be sent between the BIC ANSI Interface and the co-network has a different set of required elements. When the BIC ANSI Interface process receives an external message, it expects to receive the required elements and only the required elements for the external message type. Internally, the BIC ANSI Interface expands the compressed external message into a larger external message structure capable of holding all of the possible external message elements. This same larger structure is filled in with the information for the response message to the co-network. After all of the information for the response is added to the expanded external message, the message is compressed to contain only the required fields for that message type. The required fields for each message type are determined by a byte map, which is included as part of the external message.

The byte map starts out as 64 bits of information, where each bit signifies the absence or presence (binary 0 or 1, respectively) of a data element in the message. Sixty-four bits are converted to and from 16 bytes of display data using hexadecimal equivalents:

Conversion Table

Bit Value	Hex Value	Bit Value	Hex Value
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

Since transparent data transmission is not supported, the field is converted to display format by doing a “hexout” (i.e., each four bits of information is converted to the display form of a hex digit). In this case, 64 bits are converted to 16 bytes. Bits are numbered from left to right, starting with 1.

Byte maps shown in the following table may or may not be product-dependent. The Product column is blank for a message type with a byte map that is not product-dependent—one or more products can use this byte map. The Product column contains an entry identifying a message type with a product-dependent byte map—only the product listed can use this byte map: BASE24-atm (ATM), BASE24-pos without address verification (POS), or BASE24-pos with address verification (POS/AV). The value of the BIC-VERSION-NUM param in the Logical Network Configuration File (LCONF) determines the byte maps used for BASE24-pos 0200, 0210, 0220, and 0221 messages. If the param is set to 00000300 (Current version with address verification), the POS/AV byte maps are used. Otherwise, the POS byte maps are used.

Message Type	Product	Description	Byte Map	Message Length
0200	ATM	Financial request	E748000262000E14	278 bytes
0200	POS	Financial request	F74E000262003634	350 bytes
0200	POS/AV	Financial request	F77E000262003634	380 bytes
0210	ATM	Financial response	3C802000880000D0	142 bytes
0210	POS	Financial response	F4C02000880040D0	175 bytes
0210	POS/AV	Financial response	F4D02000880040D0	176 bytes
0220	ATM	Force post transaction advice	E7480002E80004D4	299 bytes
0220	POS	Force post transaction advice	F74E0002E80074F4	398 bytes
0220	POS/AV	Force post transaction advice	F77E0002E80074F4	428 bytes
0221	ATM	Financial completion advice	E7480002E80004D4	299 bytes
0221	POS	Financial completion advice	F74E0002E80074F4	398 bytes
0221	POS/AV	Financial completion advice	F77E0002E80074F4	428 bytes
0230		Advice acknowledgment	0400000008000010	50 bytes

Message Type	Product	Description	Byte Map	Message Length
0400	ATM	Financial reversal	3500100018000010	80 bytes
0400	POS	Financial reversal	F74E1002F80074F0	365 bytes
0402		Chargeback	F6CE2002EA0076F4	426 bytes
0410		Financial reversal acknowledgment	0400000008000010	50 bytes
0412		Chargeback acknowledgment	F6CE2002EA0076F4	426 bytes
0500		Settlement totals	050003FC00000010	132 bytes
0510		Settlement totals acknowledgment	0500000000000010	50 bytes
0700	ATM	Inquiry	C748000262000E14	268 bytes
0700	POS	Inquiry	C74E000262002614	287 bytes
0710	ATM	Inquiry response	1C802000880000D0	134 bytes
0710	POS	Inquiry response	5C802002880000D0	142 bytes
0800		Network management message	0580000101000100	65 bytes
0800		Dynamic key management/key management	0580000101008100	275 bytes
5400		Financial adjustment	F7482002E80004D4	317 bytes

In addition to the data elements identified by the byte map, each message includes two additional fields: the TYP and the BYTE-MAP fields. These fields contain the message type and the byte map for the message. They are required for every message exchanged between the BIC ANSI Interface and the co-network.

BIC ANSI External Message

The following pages describe the expanded BIC external message. Descriptions are provided for all of the fields that may be included in the variable external message. Each description identifies the data element, bit map indicator, and field length.

Each boxed field or set of fields corresponds to one bit map position. In some cases, different information is identified by the same bit map position. In those cases, each field or group of fields identified by the bit map position is described separately.

Element	Bit	Length	Field Name and Description	Data Type
1	None	4	TYP The message type. Valid values are as follows: 0200 = Financial request 0210 = Financial response 0220 = Force post transaction advice 0221 = Financial completion advice 0230 = Advice acknowledgment 0400 = Financial reversal 0402 = Chargeback 0410 = Financial reversal acknowledgment 0412 = Chargeback acknowledgment 0500 = Settlement totals 0510 = Settlement totals acknowledgment 0700 = Inquiry 0710 = Inquiry response 0800 = Network management message 5400 = Financial adjustment This field is required for all messages.	PIC 9(4)
2	None	16	BYTE-MAP The byte map for this message type. This field is required for all messages.	PIC X(16)

Element	Bit	Length	Field Name and Description	Data Type
3	1	2	PAN.LGTH The actual length of account number in the PAN.NUM field.	PIC 9(2)
4	None	19	PAN.NUM The card number (PAN) identifying the card. The value in this field should be left-justified and blank-filled.	PIC X(19)
5	2	6	TRAN-CDE The following fields contain transaction information.	
		2	TYP The transaction type.	PIC X(2)
		2	FROM-ACCT-TYP The account type of the <i>from</i> account for this transaction.	PIC X(2)
		2	TO-ACCT-TYP The account type of the <i>to</i> account for this transaction.	PIC X(2)
		6	POS-TRAN-CDE The following fields contain BASE24-pos transaction information.	REDEFINES TRAN-CDE
		2	TYP The transaction type.	PIC X(2)
		1	CARD-TYP The card type.	PIC X

Element	Bit	Length	Field Name and Description	Data Type
		2	ACCT-TYP The account type.	PIC X(2)
		1	CATEGORY The transaction category.	PIC X
6	3	8	AMT-1 The transaction amount.	PIC 9(8)
7	4	8	AMT-2 The value in this field depends on the type of transaction. For a BASE24-pos purchase with cash back transaction request, this field contains the <i>cash back</i> amount. For a financial transaction response, it contains a balance amount. If the amount is negative, position zero (0) contains a minus sign.	PIC 9(8)
8	5	8	AMT-3 For 0210 messages this field contains a balance amount. If the amount is negative, position zero (0) contains a minus sign.	PIC 9(8)
9	6	6	ACQ-TRACE-NUM The acquirer trace number assigned to this transaction.	PIC 9(6)
10	7	6*	TRAN-DAT The date of this transaction (yymmdd). *For 0700 messages, this field is only 4 bytes long (MMDD).	PIC X(6)

Element	Bit	Length	Field Name and Description	Data Type
11	8	6	ACQ-POST-DAT The posting date for this transaction according to the transaction acquirer. In settlement messages, this is the settlement date to which the totals apply. In 0800 cutover and cutover acknowledgment messages, this is the date of the day being cutover (ended).	PIC X(6)
12	9	6	RESP-ACQ-POST-DAT For 0210 and 0710 response messages, the posting date for this transaction according to the transaction acquirer. For 0800 cutover and cutover acknowledgment messages, this is the next settlement date (new day being started).	PIC X(6)
13	10	6	TRAN-TIM The following fields contain the time of the transaction. TIM-OFST must be added to this time in order to determine the local terminal (receipt) time.	
		2	HH	PIC 99
		2	MM	PIC 99
		2	SS	PIC 99
14	11	29	ADDR-VER For BASE24-pos 0200, 0220, and 0221 messages with address verification, the following fields contain cardholder information received with the transaction.	
		20	ADDR The cardholder billing address.	PIC X(20)

Element	Bit	Length	Field Name and Description	Data Type
		9	ZIP-CDE The cardholder billing ZIP code. Five-digit ZIP codes must be left-justified. Remaining positions must be blank.	PIC X(9)
15	12	1	ADDR-VRFY-STAT For BASE24-pos messages with address verification, a code identifying the result of comparing address verification information received in the transaction and address verification information contained in the processor's data base. Valid values are as follows: A = Address — Addresses matched, but ZIP codes did not match E = Error — The transaction was not eligible for address verification or an editing error occurred while attempting to process the message N = No — Addresses did not match and ZIP codes did not match R = Retry — Primary and secondary authorizers were unavailable or have declined the transaction and address verification is not performed on BASE24-pos S = Service not supported — BASE24-pos authorized the transaction, but did not have the add-on Address Verification module U = Unavailable — Address information was not available to the processor performing address verification W = Whole ZIP — Nine-digit ZIP codes matched, but addresses did not match X = Exact — Addresses and nine-digit ZIP codes matched Y = Yes — Addresses and five-digit ZIP codes matched Z = ZIP — Five-digit ZIP codes matched, but addresses did not match	PIC X
16	13	10	ACQ-INST-ID-NUM The routing and transit number of the terminal-owning financial institution.	PIC 9(10)

Element	Bit	Length	Field Name and Description	Data Type
17	14	10	CRD-ACCPT-ID-NUM The routing and transit number of the card-accepting financial institution, if the acquirer was not the actual card acceptor.	PIC 9(10)
18	15	3	TERM-CNTRY-CDE The country code of the terminal-owning institution.	PIC 9(3)
22	19	10	ISS-INST-ID-NUM The routing and transit number of the card-issuing financial institution.	PIC 9(10)
23	20	2	RVSL-CDE For 0400 reversal messages, the reason for the reversal. Valid codes are as follows: 01 = Time-out 02 = Command reject (invalid response) 03 = Destination not available 08 = Customer canceled 10 = Hardware error 21 = MAC failure 22 = KMAC synchronization error 23 = Message replay 24 = Invalid MAC	PIC 9(2)
26	23	11	GROSS-DB-AMT The gross debit amount, including reversal transactions.	PIC 9(11)
27	24	10	GROSS-DB-CNT The gross debit count, including reversal transactions.	PIC 9(10)
28	25	11	GROSS-CR-AMT The gross credit amount, including reversal transactions.	PIC 9(11)

Element	Bit	Length	Field Name and Description	Data Type
29	26	10	GROSS-CR-CNT The gross credit count, including reversal transactions.	PIC 9(10)
30	27	10	RVSL-DB-AMT The reversal debit amount.	PIC 9(10)
31	28	10	RVSL-DB-CNT The reversal debit count.	PIC 9(10)
32	29	10	RVSL-CR-AMT The reversal credit amount.	PIC 9(10)
33	30	10	RVSL-CR-CNT The reversal credit count.	PIC 9(10)
34	31	2	PT-TRAN-SPEC A point-of-transaction special condition code. A 03 in this field indicates an ATM transaction. Any other value indicates a POS transaction. This field is also used to carry the adjustment code (reversal code) for an ATM adjustment transaction (5400).	PIC 99
35	32	3	NMM-INFO-CDE The network management information code. Valid values are as follows: 001 = Positive logon response 002 = Logoff 003 = Negative logon response (system or line not available) 103 = Negative logon response (password/options incorrect) 161 = Change PIN key 162 = New PIN key 163 = Repeat PIN key	PIC X(3)

Element	Bit	Length	Field Name and Description	Data Type
			164 = Verify PIN key 201 = Initiate cutover 261 = Change PIN key positive response 262 = New PIN key positive response 263 = Repeat PIN key positive response 264 = Verify PIN key positive response 301 = Logon (echo test) 361 = Change PIN key negative response 362 = New PIN key negative response 363 = Repeat PIN key negative response 364 = Verify PIN key negative response 601 = Cutover acknowledgment	
36	33	8	RESP-CDE-R	PIC X(8)
		8	RESP-CDE	REDEFINES RESP-CDE-R
		5	FILLER	PIC X(5)
		3	RESP	PIC X(3)
			This field contains a 3-digit code in the format “crr”, where <i>c</i> indicates the card disposition and <i>rr</i> is the reason code. Valid values for <i>c</i> are as follows: 0 = Return the card 1 = Retain the card	
		8	ORIG-AMT	REDEFINES RESP-CDE-R
			The original amount for adjustment transactions.	

Element	Bit	Length	Field Name and Description	Data Type
37	34	25	TERM-OWNER-NAME The name of the financial institution that owns the terminal at which this transaction was initiated.	PIC X(25)
38	35	15	TERM-CITY-ST The transaction-initiating terminal's city and state.	PIC X(15)
39	36	6	ORIG-DATA.ORIG-ACQ-POST-DAT The acquirer network's posting date for the original transaction. This date allows the issuer interface module to know which log file to access in order to find the original transaction for this interface.	PIC 9(6)
40	37	6	ISS-TRACE-NUM The trace number assigned to this transaction by the issuer interface.	PIC 9(6)
42	39	16	PIN The entered PIN.	PIC X(16)
43	40	8	PASSWORD The version number of the BIC ANSI Interface. This is a 3-byte field right-justified and zero-filled to the left. Valid values are as follows: 00000100 = Previous release 00000200 = Current release 00000300 = Current release with address verification	PIC X(8)

Element	Bit	Length	Field Name and Description	Data Type
52	49	208	KEY-MGMT This field consists of the following information:	PIC X(208)
		2	KEY-DIR Valid values are as follows: 00 = Both 01 = Outbound 02 = Inbound	
		6 200	CHK-DIGITS 917-CSM-DATA	
53	50	8	APPRV-CDE The approval code for a POS transaction.	PIC 9(8)
54	51	25	POS-INFO The following fields are used by BASE24 to properly track transactions and perform authorization and settlement processing for POS transactions.	
		1	DFT-CAPTURE-FLG An indicator of whether this transaction is a draft capture item.	PIC 9
		1	SETL-FLG Not Used	PIC 9
		10	INVOICE-NUM The invoice number associated with this transaction.	PIC X(10)

Element	Bit	Length	Field Name and Description	Data Type
		10	ORIG-INVOICE-NUM The original transaction invoice number associated with this transaction.	PIC X(10)
		3	PT-SRV-ENTRY-MDE A code indicating the manner in which this transaction entered BASE24-pos.	PIC 9(3)
55	52	30	ISO-ORIG-DATA The following fields contain ISO-defined original data elements. They are included in reversal and adjustment transactions to identify the original transaction being reversed or adjusted.	
		4	MSG-TYP The transaction type identifying the original transaction.	PIC X(4)
		6	TRAN-TIM The time of the original transaction.	PIC 9(6)
		4	TRAN-DAT The date of the original transaction.	PIC X(4)
		12	SEQ-NUM The sequence number of the original transaction.	PIC X(12)
		4	B24-POST-DAT The date the original transaction was posted by BASE24.	PIC X(4)

Element	Bit	Length	Field Name and Description	Data Type
56	53	19	SHRG The following fields are used by BASE24 to determine if terminal sharing is allowed for a transaction. The first field is the terminal county code and the second is a table of values for every transaction class supported by BASE24, the values indicating whether the transaction class can be shared within the county, between counties, between states, or internationally for the terminal.	
		3	TERM-CNTY	PIC 9(3)
		16	NOT-ON-US-MASK	PIC X(16)
57	54	51	TERM The following fields contain information about the transaction-initiating terminal.	
		25	TERM-NAME-LOC The terminal location.	PIC X(25)
		16	TERM-ID The terminal identification.	PIC X(16)
		6	SEQ-NUM The sequence number associated with this transaction by the terminal.	PIC 9(6)
		4	TIM-OFST The time difference between the transaction-initiating terminal and the Tandem processor location. It is the signed number of minutes to be added to the acquirer Tandem time in order to obtain the terminal local time.	PIC S999

Element	Bit	Length	Field Name and Description	Data Type
58	55	2	PIN-SIZE The number of digits in the PIN or in the PIN block. If no encryption is used, this field reflects the number of digits in the entered PIN. If the PIN is encrypted, this field reflects the number of digits in the PIN block. PIN blocks contain 16 digits.	PIC X(2)
59	56	16	OPTIONS The configuration options. Each byte represents one configuration option. The table below identifies the option associated with each byte. 1 = Completion (0221) messages 2 = Completion acknowledgment (0230) messages 3 = Stand-in Authorization and Force-Post (0221) messages 4 = Reversal acknowledgment (0410) suppression 5 = Reversal (0400) after Completion (0221) suppression 6 = Pin Encryption type. Valid values are as follows: 0 = Clear 1 = Security module encryption 2 = DES software encryption 7 = Cutover option. Valid values are as follows: 0 = Equal Partner 1 = Master 2 = Slave 8 = Number of Keys 9 = Key processor type Except where otherwise noted, valid values for each position are as follows: 0 = Not supported 1 = Supported and required Note that for bytes 4 and 5, a value of 0 means that suppression is not supported.	PIC 9(16)
60	57	6	ISS-POST-DAT The date of this transaction for the issuer network.	PIC X(6)

Element	Bit	Length	Field Name and Description	Data Type
61	58	38	ACCT-NUM The following fields contain account numbers affected by this transaction.	
		19	FROM-ACCT The <i>from</i> account number of the transaction.	PIC X(19)
		19	TO-ACCT The <i>to</i> account number of the transaction.	PIC X(19)
		38	POS-DATA The following fields are used to describe the account used in POS transactions.	REDEFINES ACCT-NUM
		19	ACCT The account number.	PIC X(19)
		3	MBR-NUM The member number of the card used to initiate this transaction. When multiple cards are issued with the same card number, this field distinguishes between cards.	PIC X(3)
		2	RSN-CDE	PIC X(2)
		2	SERVICE	PIC X(2)
		2	REA-FOR-CHRGBCK	PIC X(2)
		1	NUM-OF-CHRGBCK	PIC X
		9	USER FIELD 2	PIC X(9)

Element	Bit	Length	Field Name and Description	Data Type
62	59	15	PRE-AUTH The following fields contain preauthorization information.	
		12	SEQ-NUM The sequence number of the transaction.	PIC X(12)
		3	HLD The preauthorization hold time or life cycle of a preauthorization transaction. The hold time is expressed in number of minutes to hold funds, and will contain zeros on all transactions which are not preauthorization transactions.	PIC 9(3)
63	60	2	ADD-DATA-LGTH Not Used	PIC 9(2)
64	None	16	ADD-DATA Not Used	PIC X(16)
		2	TRACK2.LGTH The length of the Track 2 data.	PIC 99
68	None	39	TRACK2.T2DATA The actual Track 2 data, starting with the first byte after the start sentinel.	PIC X(39)

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